The Economic Incidence of Natural Disasters

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Abstract

Policymakers and researchers often view the costs of natural disasters as local to the areas they physically strike. However, we show that a disaster's economic incidence differs substantially from its physical footprint. We study Pakistan's large and unforeseen 2022 floods, using original panel surveys of 5,100 households, satellite flood maps and a recentered IV design based on a hydrological model. A 10th-to-90th percentile increase in flooding within 15 km of a home reduces real expenditure by 10%, and nearly two-thirds of this decline is due to flooding more than 5 km away. The same spatial pattern holds for men's wage employment, prices, asset sales and credit access. Commuting adjustments help explain how these spillovers arise. Holding surrounding flooding constant, more flooding within 5 km increases the likelihood of commuting further for work by 4 percentage points, while flooding 5–15 km away offsets that increase. This suggests that households rely on nearby markets to cope with local losses, but lose that option when flooding becomes widespread. We quantify the incidence of flooding in general equilibrium by building a spatial model of trade and commuting. The model implies that whilst 40% of locations experienced almost no flooding, welfare fell everywhere, and 73% of the loss was due to flooding elsewhere. Targeting aid on model-predicted losses rather than physical exposure achieves the same welfare at 18% lower cost. Thus, ignoring the divergence between *incidence* and *physical exposure* leads us to underestimate disaster costs and misallocate aid.

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1 INTRODUCTION

Governments, aid agencies and researchers often view the impacts of natural disasters as local, assuming that those directly hit bear most of the costs. Yet spatial linkages can transmit losses far beyond the locations a disaster physically strikes. Flooded households, for instance, may buy goods and find work in drier areas. This provides them with partial insurance against their own losses but shifts some of the costs onto residents elsewhere through changes in prices and wages. If these spillovers are large, ignoring them will lead us to misjudge both the size and the distribution of the impacts of natural disasters, and to misallocate aid.

This paper estimates the economic incidence of Pakistan's 2022 floods in general equilibrium. These floods were the largest in the country's 75-year history and were induced by unprecedented rainfall. In Sindh, the hardest-hit province, summer rain was 267% above its long-run average. Nationwide, the floods inundated 7.3% of Pakistan's land and directly affected 14% of its population. Globally, these floods rank in the top 5% by share of land inundated and the top 1% by share of population affected, compared to other major floods in the past quarter century.

In this setting, we first show that, empirically, the economic incidence of flooding differs substantially from its physical footprint. To do so, we draw a random sample of 5,100 low-income rural households from a pre-flood census covering six districts in Sindh. We then locate, follow and survey these households over the two years following the floods. We combine our survey data with satellite flood maps and a novel research design using recentered instruments (Borusyak and Hull, 2023, 2026). The design exploits deviations between actual 2022 flooding and expected flooding at each location, where the latter is simulated using a hydrological model and satellite data on historical rainfall.

We then build a spatial model to quantify the general-equilibrium effects of flooding. To assess how broadly our findings generalise, we also simulate counterfactual floods of different magnitudes and under different levels of market integration. Finally, we use the model to study how spillovers shape the efficient allocation of disaster aid, allowing for the possibility that aid itself generates general-equilibrium effects.

Our instrumental-variable (IV) estimates show that a household's expenditure loss depends more on flooding in the surrounding area than on flooding at its own location. We find that real expenditure declines by 10.1% ($p = 0.01$) when we move a household from the 10th to the 90th percentile of flooding in a 15 km radius around its house.[1] Since our surveyed households vary considerably both

[1] We limit ourselves to 15 km due to a loss of statistical power beyond 15 km.

in their direct, physical exposure to flooding and, conditional on that, flooding in the wider area, we can decompose this total effect over 15 km into the contribution of nearer versus more distant flooding. For this exercise, we add separate regressors for flooding within c km and flooding in the outer c-to-15 km ring. This exercise shows that flooding within 1 km of home accounts for only about 11% of the expenditure decline, while flooding more than 5 km away accounts for 64%.

Why does flooding several kilometres from a household affect its consumption? We find that the answer lies in linkages through labour, product and credit markets across space. In theory, under autarky only flooding at a household's own location would matter. In practice, our results show that access to nearby locations helps households cope with local losses, unless those locations are also flooded. By the same token, households can lose when nearby locations are flooded, even without being flooded themselves.

Commuting illustrates this pattern well. Holding flooding in other distance bands constant, flooding within 5 km raises commuting by about 4 percentage points. By contrast, flooding 5–15 km away lowers it enough to offset that increase. In other words, the commuting response to localised flooding is offset when flooding also affects the surrounding area. This is consistent with household members seeking work in nearby places after local flooding, while widespread flooding reduces those opportunities.

Employment shows a similar pattern. A 10th-to-90th percentile increase in flooding within 15 km of a house reduces men's wage employment by 9 percentage points. Over the same radius, the food price index rises by 8%, the share of households with loans falls by 5 percentage points, and asset sales rise by about 3 percentage points. Flooding beyond 1 km accounts for nearly all of the price and loan effects and most of the employment and sales results. Together, these findings suggest that spatial links can help households cope with localised flooding, while also spreading the costs of flooding across the region.

A potential concern with our analysis is that measurement error in household locations could inflate our estimates of spillovers. If a household is recorded in the wrong place, distant flooding may proxy for local flooding even after controlling for our measure of it. We test for this using housing damage as a placebo outcome. Our measure comes from a geo-tagged census of properties that the Government of Sindh conducted after the floods to assess damage and guide reconstruction. Housing damage is a useful placebo because a home cannot be damaged by floodwaters far away, so any estimated effect of distant flooding on damage would suggest measurement error. We find no such effect: flooding within 1 km raises the probability of damage by 6 percentage points, while flooding further away has no detectable effect.

Our IV estimates rely on cross-sectional variation in flooding, so they cannot recover the full general-equilibrium effects of the flood. Comparisons between more- and less-flooded households net out impacts which are common to both, a familiar problem for those seeking to learn about aggregate effects from regional variation (e.g. Nakamura and Steinsson, 2014; Chodorow-Reich, 2019; Autor et al., 2013). With spatial frictions, the missing effects are not a single intercept but differ across locations (Adão et al., 2026). We must therefore recover these general-equilibrium effects to measure both the aggregate impact of the flood and the spatial distribution of total losses.

To quantify the total effect of the flood on welfare in each location, we use a spatial model in which locations are linked through costly trade and commuting, based on Eaton and Kortum (2002) and Ahlfeldt et al. (2015). There are two sectors of production, agriculture, which uses labour and land, and non-agriculture, which only uses labour. Consistent with our data, workers live in a fixed location and choose where and in which sector to work. Worker welfare depends on access to well-paying jobs, cheap goods and their housing endowment. Flooding lowers agricultural productivity and destroys housing in affected locations. As workers adjust where they buy and work, prices and wages change throughout the economy. The extent of these spillovers depends on pre-flood economic integration and on how readily buyers and workers can substitute across locations.

We discipline these links using data from our setting, starting with bilateral travel times computed from the road network. We use internal trade data to estimate how trade flows vary with travel time, then separately identify the trade elasticity by regressing post-flood prices on flood-induced changes in market access using a recentered instrument. We estimate how commute flows vary with travel time using data from our surveys, and calibrate the commuting elasticity to match the observed dispersion of wages.

We quantify the model for the province of Sindh, which we divide into approximately 13,400 hexagons, each with a radius of 1.2 km. Using data on agricultural revenues, population and agricultural employment, we recover pre-flood productivity in each location. We then combine the flood map with the estimated effects of flooding on agricultural output and housing damage to compute post-flood fundamentals, and solve for the resulting equilibrium. This approach allows us to map realised or counterfactual floods into welfare losses across the entire province.

The model results show that the economic impacts of the 2022 floods extended well beyond the locations they inundated. Two in five locations experienced almost no flooding, yet welfare fell in every location, by at least 1.6% and by 12.2% on average. For the average hexagon, 73% of the welfare loss was caused by flooding in other hexagons. This share was 99% in the least-flooded decile and 32% in the most-flooded decile.

We use model counterfactuals to show how the distribution of losses changes with a disaster's scale and spatial pattern. When 5% of locations are flooded, the welfare loss per resident in a flooded location is 3.8 times that in an unflooded location. When 60% are flooded, this ratio falls to 2.1, as losses in unflooded locations nearly triple. Dispersing flooding across space raises the share borne by unflooded locations from 24% to 30%, whilst leaving the aggregate loss almost unchanged.

We also vary market integration to assess how trade and commuting links affect the reach of these losses. These counterfactuals clarify why comparisons between more- and less-exposed households measure relative effects: when flooding affects outcomes in both groups, the comparison nets out part of the total impact. This gap grows with market integration and with flood size over the range we study, attenuating the estimated flood impacts.

The distinction between economic incidence and physical exposure matters for the allocation of disaster aid. A rank-rank regression of model-predicted welfare losses on own flooding has an R^2 of 0.71, so physical exposure predicts losses but does not fully capture them. As a result, a rule that transfers aid to every flooded location and one that targets an equal number of locations selected by predicted welfare loss disagree for one in six locations. Those missed by the exposure-based rule tend to be less-flooded locations surrounded by extensive flooding, for which flooding elsewhere causes 92% of the loss.

To value the potential gains from targeting aid according to economic incidence we introduce a planner who allocates a fixed budget to minimise a convex function of flood-induced welfare losses. Compared to targeting transfers proportional to a location's own flooding, targeting on model-predicted losses could achieve the same welfare at 82% of the cost. These savings depend on the planner's ability to target losses directly, and fall to 3% if transfers can be conditioned only on own flooding. These calculations assume that aid itself has no general-equilibrium effects, which we then relax, allowing aid that restores productive capacity to change wages and prices through the same spatial links that transmit flood losses.

We contribute primarily to the literature on the impacts of natural disasters and extreme weather events, in two ways.

First, we quantify the incidence of a disaster, the distribution of its costs across space, in general equilibrium. Existing work documents direct effects of disasters on households by comparing affected and unaffected households or locations (Hornbeck, 2012; Hornbeck and Naidu, 2014; Gignoux and Menéndez, 2016; Gröger and Zylberberg, 2016; Gallagher and Hartley, 2017; Kirchberger, 2017; Deryugina et al., 2018; Boustan et al., 2020; Groen et al., 2020; Kocornik-Mina et al., 2020; Patel, 2024; Baseler and Hennig, 2026), or traces specific spillover channels, such as supply-chain

linkages (Barrot and Sauvagnat, 2016; Boehm et al., 2019; Elliott et al., 2019; Carvalho et al., 2021; Castro-Vincenzi, 2024; Pankratz and Schiller, 2024; Balboni et al., 2026). These studies leave open the question of how the total cost of a disaster is distributed, which matters for how we quantify the losses from disasters, assess exposure to climate risk and target aid. We distinguish the effects of direct and indirect exposure on the same households, using a spatial spillover design similar to field experiments that estimate spillovers from the treatment of neighbouring units (Miguel and Kremer, 2004; Egger et al., 2022; Muralidharan et al., 2023). Because these relative estimates miss general-equilibrium effects, which with spatial frictions differ across places (Adão et al., 2026), we recover the spatial distribution of a disaster's costs using a quantitative spatial model.[2]

Second, we show how the incidence of a disaster depends on the number of places it shocks simultaneously. A literature in macroeconomics uses cross-sectional variation to estimate the impact of a shock, such as an increase in government spending, on a single region or household, and asks how this relates to the impact of the same shock applied to all units simultaneously (Nakamura and Steinsson, 2014; Chodorow-Reich, 2019, 2020; Guren et al., 2021; Wolf, 2023). Chodorow-Reich (2019, 2020) argues that the two differ because relative estimates difference out general-equilibrium effects common to all regions, and because trade and factor mobility change the effect of a shock on the region it hits. Work on scaling up interventions similarly emphasises that estimates from small-scale programmes may not carry over to large-scale ones, because of general-equilibrium effects (Heckman et al., 1998; Acemoglu, 2010; Crépon et al., 2013; Banerjee et al., 2017; Muralidharan and Niehaus, 2017; Bergquist et al., 2026). One implication of this work is that the incidence of a shock depends on the share of units it hits. We build on this insight, showing how the spatial incidence of a flood changes with the share of places flooded. We find that indirect exposure matters more than direct exposure in the 2022 floods, and our counterfactuals suggest that this is in part because they flooded so many places at once. This has two implications for the study of disasters. One is that relative estimates are likely to better capture the losses of directly affected places in small disasters, which the literature more often studies, than in large ones. The other is that estimates from past events may understate the costs of the large-scale events that climate change is making more common, adding to concerns about using historical experience to project the damages from climate change (Dell et al., 2014).

Our results also speak to work on how households are insured against local shocks, through economic integration (Newbery and Stiglitz, 1984; Jayachandran, 2006; Burgess and Donaldson, 2010; Brooks and Donovan, 2020; Caselli et al., 2020; Allen and Atkin, 2022) and informal risk-sharing

[2] Donaldson et al. (2025) use a related approach to answer a different question, correcting for spillovers through production networks to estimate the aggregate cost of an earthquake.

(Rosenzweig and Stark, 1989; Townsend, 1994; Jack and Suri, 2014; Munshi and Rosenzweig, 2016; Morten, 2019; Kinnan et al., 2024). Markets spread the costs of a disaster across space, and so partly insure households against flooding at home, though less so when a disaster covers more of the places around them. Although we study a single event, this raises the possibility that market linkages, like some public policies (Annan and Schlenker, 2015; Fried, 2022; Baylis and Boomhower, 2023; Balboni, 2025; Hsiao, 2025), crowd out private efforts to reduce physical exposure to climate risk.

Finally, we contribute to the literature on the targeting of social programmes, which studies how best to allocate resources when the planner faces information constraints (Ravallion, 1993; Elbers et al., 2007; Alatas et al., 2012, 2016b,a; Hanna and Olken, 2018; Aiken et al., 2022; Banerjee et al., 2024; Hanna et al., 2026). In our context, the planner uses transfers to offset the impact of floods. The losses from a flood are causal effects, determined in general equilibrium, and are not directly observable. Our problem therefore relates closely to work on targeting transfers by their estimated treatment effects (Manski, 2004; Bhattacharya and Dupas, 2012; Kitagawa and Tetenov, 2018; Athey and Wager, 2021; Haushofer et al., 2025; Ernst et al., 2026), and to the literature on optimal spatial transfers, which uses a model-based approach to measure the general-equilibrium impact of transfers on economic geography (Fajgelbaum et al., 2019; Fajgelbaum and Gaubert, 2020, 2025; Gaubert et al., 2025; Donald et al., 2026; Rossi-Hansberg et al., 2026). We develop a feasible approach to targeting aid on these losses, using a model quantified with widely available data, which can also account for the general-equilibrium effects of the transfers themselves. In the context of disasters, Gordon et al. (2025) find that property damage is a poor basis for targeting reconstruction aid, due to household-level heterogeneity in access to informal insurance. We show that physical exposure is also an imperfect basis for targeting, for a different reason, since it misses losses that propagate across space through markets.

The rest of the paper proceeds as follows. Section 2 describes the 2022 floods and Section 3 the data. Section 4 sets out the research design and Section 5 presents the reduced-form results. Section 6 develops, estimates and quantifies the spatial model. Section 7 evaluates the incidence of the 2022 floods and of counterfactual floods, and Section 8 studies the targeting of disaster assistance. Section 9 concludes.

2 PAKISTAN'S 2022 FLOODS

Our setting is Pakistan's 2022 floods, which were the worst in the country's 75-year history.

Heavy rain is not unusual in Pakistan, which experiences a monsoon every summer. However, in 2022 the cumulative July–August rainfall was the greatest it had been in four decades – 114%

above the 1981–2021 average, and 27% above the previous wettest year (2010) (Funk et al., 2015). The rain was concentrated in Sindh, the southernmost province and our study region, where it was 267% above the long-run average, and 44% above its own previous record (2003). It is very likely that climate change contributed to this unprecedented event. An attribution study by World Weather Attribution estimated that the rainfall was 75% more intense than it would have been without human-induced global warming (World Weather Attribution, 2022).

The rains caused widespread flooding. According to MODIS satellite imagery (Nigro et al., 2014), 7.3% of normally dry land in Pakistan was inundated at some point between July and December 2022. In Sindh the figure was 20.4%, and in the six districts where we work it reached 41%. Because Sindh is low-lying and flat, the water had nowhere to drain, and large areas of the province stayed inundated for months whilst the waters slowly evaporated.

The floods were extreme by global standards, not only Pakistan's. The Global Flood Database of Tellman et al. (2021) collates information on the 906 largest floods worldwide since 2000, taking as its starting point the Dartmouth Flood Observatory's log of floods reported in news and government sources. For each flood, it uses satellite imagery to record the share of national land and population that was inundated. Across these floods, a flood at the 90th percentile inundated 2.8% of national land and directly affected 1.7% of national population. Pakistan's 2022 floods covered 7.3% of the country's land and directly affected 14% of its people, placing this event at the 95th and 99th percentiles, respectively.[3]

This large-scale event is a compelling setting for our research question. In part this is a matter of power, since the larger the event, the more likely we are to detect general-equilibrium impacts. But more importantly, enormous floods are worth understanding in their own right, since it is likely that more of the floods we face in coming years will be large ones. Climate change is making weather extremes more frequent, and a single event increasingly covers more of a country at once.[4]

3 DATA

To estimate how household outcomes respond to flooding at different distances, we need geolocated data on consumption, employment and prices. Data of this kind is hard to come by in Pakistan.

[3] It is clear from Patel (2024) that many floods, particularly smaller ones, are omitted from the DFO. This suggests we may be underestimating how extreme Pakistan's 2022 floods were in the global context.

[4] The area a single event covers has widened across hazards, most clearly for heat (Skinner et al., 2025) and drought (Brunner and Chartier-Rescan, 2024), and also for tropical-cyclone rainfall (Qin et al., 2024) and European floods (Kemter et al., 2020). In South Asia, both widespread monsoon rainfall (Roxy et al., 2017) and large-scale heat extremes (Banerjee et al., 2025) have become more common.

Administrative data has limited coverage or is not regularly updated. The government suspended its household surveys for census preparations, and those it does field ask few of the questions we need. Satellite data does not record prices, consumption or incomes.

We therefore bring together data from a wide range of sources with granular geographic information, down to points or hexagons a kilometre across. Our principal source is an original, geocoded panel of rural households, which we collected to measure household outcomes at the level our question requires. To measure physical damage, we combine official microdata on housing damage with satellite building footprints. For agricultural output and revenues, we classify plots by crop type using satellite data and machine learning, proxy output with NDVI, a satellite index of greenness, calibrate to district-level agricultural surveys, and convert to revenues using prices from major markets. We measure flooding from multiple satellite sources.

This section also describes the topographical and rainfall data underlying our identification strategy, and the internal trade, transport network, population and labour force data we use to quantify the model. Table 3.1 summarises the sources.

Table 3.1: Data sources

Variable	Source	Years	Resolution
Household survey			
Household outcomes	Original panel	2017, 2023, 2024	Household (GPS)
Physical damage and agriculture			
Building damage rate	Damage census; VIDA	2022–23	Hex, 460 m radius (res. 8)
Agricultural revenues	Sentinel-1/2; MODIS; SBOS; SPI	2018–2024	Hex, 1.2 km radius (res. 7)
Agricultural employment share	Damage census; PLFS	2020–23	Hex, 460 m radius (res. 8)
Flooding			
Flood index	MODIS; Sentinel-1	Jul–Dec 2022	Hex, 170 m radius (res. 9)
Hydrological model inputs			
Rainfall	IMERG; ECMWF forecasts	1998–2025	0.1°–0.4°, daily
Topography	FABDEM	Static	30 m
Model inputs			
Trade flows	Market arrival records	2018–2026	Market, daily
Road network	OpenStreetMap	Static	Road segment
Population	WorldPop	2020	100 m
Housing stock	GHSL built-up surface	2020	100 m

3.1 Household survey data

We collect panel data on 5,100 poor, rural households in Sindh, first observed in 2017 and surveyed twice after the floods, in summer 2023 and 2024.

We draw the sample from a census of all low-income households in six districts of the province (Qambar Shahdadkot, Larkana, Matiari, Tando Muhammad Khan, Tando Allahyar and Sujawal). Whilst restricting attention to the rural poor limits our ability to comment on the experience of wealthier or urban households, its key benefit is that we can compare very similar households that happened to face very different flooding.

A local NGO collected the data to determine eligibility for development and poverty alleviation programmes. It contains the address of the household, a list of household members (including their age, sex, education and occupation), household asset holdings across 21 asset classes and a poverty score constructed from these asset holdings. In addition to treating this data as a sampling frame, we extract baseline covariates from it, which we use as controls to improve precision and to run balance checks.

We sample to span the range of flood exposure, choosing survey locations within each district with high and low predicted flooding, and then randomly drawing households from them. We survey households in two waves, running July–September of 2023 and 2024. With the assistance of local community members, we locate 92% of households in 2023 or 2024; of those, almost all are still living in their 2017 village.[5] Attrition is low and does not vary with flood exposure. In total we have 5,120 households and 33,202 people in 2023, and 4,902 households and 33,645 people in 2024. We conduct our main analysis on the 4,683 households surveyed in both waves.

The survey is extensive, so we highlight the modules that give us our main outcomes. Total expenditure and consumer prices come from a consumption module, which records total spending alongside the quantity and value consumed of meat, dairy, fruit and legumes. Employment status, occupation and wages come from a household roster, which covers demographics, education and labour for every member. Receipt of flood relief comes from a relief module, which records the type and provider of assistance alongside self-reported flood exposure and evacuation decisions. We also collect commuting data, which we use to estimate and calibrate the commuting side of the model. In the first wave we record each worker's travel time to work. In the second we additionally record their specific place of work, monetary commuting costs and commute mode.

[5]We had expected substantial permanent migration, but did not find this to be the case in our sample. Anecdotally, households expected reconstruction aid conditional on staying in place, and the worst-hit areas are not normally flood-prone, leaving little reason to relocate to reduce future risk.

3.2 Administrative damage data

To measure physical damage to housing we use the Government of Sindh's in-person census of damaged residential property, which it conducted after the floods to administer reconstruction grants, and which records 2.1 million properties. To build a damage rate, we aggregate the census to hexagons of radius ~460 m (resolution 8 of Uber's H3 system) and divide by the buildings counted there in the VIDA footprint layer, which combines the Google and Microsoft open building footprints.

The census also records whether the main earner of the household works in agriculture, which we combine with the Pakistan Labour Force Survey (PLFS) to measure the agricultural share of employment in each hexagon, which the model needs. For rural hexagons we average the microdata by hexagon, then shift every hexagon in a district by a common amount so that the district average matches the PLFS rural share. For urban hexagons we use the PLFS urban share for that district.

3.3 Agricultural revenues

We measure agricultural revenue in each hexagon of radius ~1.2 km (resolution 7), by season, from 2018 to 2024, by combining satellite data with official production statistics. We first use the satellite data to establish which crops grow where, and to build a local proxy for output. We identify the cropland in each hexagon from ESA WorldCover, a 10 m land-cover map. We then classify that cropland by crop, training a random forest on plot-level data from the Asian Development Bank (ADB) Ground Truthing Survey to predict rice, cotton, sugarcane and maize from how a plot looks to radar and optical satellites across the season. Held-out accuracy is 91%. Multiplying cropped area by NDVI in September and February of each year gives us a proxy for Kharif (summer) and Rabi (winter) output. Since this proxy is measured in greenness-weighted cropped area rather than tonnes, we rescale it so that the total for each district, crop and year matches the tonnage the Sindh Bureau of Statistics reports. We then value the tonnes at crop prices from the Sensitive Price Indicator. This gives us agricultural revenue for each hexagon in each Kharif and Rabi season.

3.4 Flood data and flood index construction

We measure flood extent and duration using MODIS, an optical sensor that maps standing water at 250 m but cannot see through cloud, and Sentinel-1, a radar sensor that sees through cloud and at night but revisits each location less frequently.[6] We measure depth by combining the Sentinel-1

[6]For an in-depth discussion of the challenges of measuring floods from space, see Patel (2024).

water maps with the FLEXTH algorithm (Betterle and Salamon, 2024). Intuitively, at the edge of a flood the water surface sits at ground level, so the flood boundary reveals the height of the floodwater above sea level, and the depth at any interior point is that water level minus the ground elevation.

We build a single index of flood intensity covering three dimensions along which floods differ – extent, duration and depth. Using each source we compute the mean share of land under water in each hexagon for each month from July to December. This gives thirteen flood measures per hexagon: six monthly flood shares from each satellite, and one depth. We take their first principal component.

3.5 Hydrological model inputs

To construct our instruments we use the LISFLOOD-FP hydrological model (Bates et al., 2010; Neal et al., 2012), which takes daily rainfall and topography (a digital map of ground elevation) as inputs. For topography we use FABDEM, which strips buildings and forest canopy from the Copernicus surface model, leaving the ground surface over which water flows, including drainage channels and irrigation canals. FABDEM does not capture large-scale flood defences, but these are largely absent in our setting. We take realised rainfall from NASA's IMERG satellite product, since Pakistan has few ground weather stations, and long-range forecasts from the ECMWF seasonal ensemble.

3.6 Trade, transport cost and other model data

To estimate trade costs we use the arrivals recorded by Punjab's market committees, which report daily to the provincial government the quantity of each commodity arriving in their market and the place it came from. This gives us bilateral trade flows between markets, which we use to estimate a gravity equation. To measure travel times we take the road network from OpenStreetMap, with road class from Geofabrik and surface quality from HOTOSM. We assign each class a speed, using published maximum speeds by road type and recorded average speeds. We then compute bilateral travel times between hexagon centres with Dijkstra's algorithm. We take the population of each location from WorldPop 2020, which distributes census counts according to where satellites detect buildings, and the housing endowment from the Global Human Settlement Layer, which records the square metres of residential building in each 100 m cell.

4 RESEARCH DESIGN

4.1 Endogeneity challenge

Our objective is to estimate the causal effect of flooding on household outcomes, using cross-sectional data collected one year after the floods. We would like to estimate

$$Y_i = \beta F_i(k) + \alpha_{D(i)} + \varepsilon_i \tag{1}$$

$$Y_i = \beta_{\text{own}} F_i^{\text{own}}(c) + \beta_{\text{near}} F_i^{\text{near}}(c) + \alpha_{D(i)} + \varepsilon_i \tag{2}$$

where Y_i is an outcome for household i, $F_i(k)$ is the mean flood index over the hexagons within k km of its home, $F_i^{\text{own}}(c)$ is the mean over the hexagons within c km, $F_i^{\text{near}}(c)$ is the mean over the hexagons between c and 15 km, $\alpha_{D(i)}$ is the fixed effect of household i's district and ε_i collects unobserved determinants of the outcome. β captures the total effect of flooding within k km of the household. β_{own} captures the effect of flooding within c km of the household, holding fixed flooding from c to 15 km, and β_{near} captures the effect of flooding from c to 15 km, holding fixed flooding within c km.

OLS estimation of equations 1 and 2 will yield biased estimates if flooding is correlated with unobserved determinants of outcomes. Such correlation may arise for physical reasons, since topography shapes both flood risk and agricultural productivity, or for behavioural reasons, if households sort and invest in response to anticipated flooding.

In both cases, the bias arises because realised flooding is correlated with expected flooding, which is in turn correlated with these unobserved determinants. We therefore employ a research design that exploits only the deviation of a location's flooding in 2022 from its expected flooding.

4.2 IV approach

We use the recentered instrumental-variable approach of Borusyak and Hull (2023), which is designed for settings in which bias arises from non-random exposure to exogenous shocks.

In our setting, the shocks we exploit are rainfall, and a location's exposure to them depends on the physical characteristics that govern how water flows and accumulates across space. Flooding in one location can therefore depend on rainfall in many other locations, since rain that falls far away but upstream of a location eventually flows to it and contributes to flooding there.

In stylised form, we write flooding in hexagon j on day t as

$$W_{jt} = w_{jt}(\mathbf{r}_{1:t};\, \mathbf{X}) \tag{3}$$

where $\mathbf{r}_{1:t}$ is rainfall in every location up to day t and $\mathbf{X}$ includes the characteristics of all locations that are fixed before the monsoon begins, observed or not, such as topography, drainage infrastructure and pre-monsoon water levels. We summarise daily flooding over the T days of the monsoon with a single index,

$$Flood_j = s(W_{j1}, \ldots, W_{jT}) \tag{4}$$

which therefore depends on rainfall in all locations on all days, $\mathbf{r} \equiv \mathbf{r}_{1:T}$.

Average rainfall varies across locations due to permanent features such as latitude, distance from the ocean and topography, and across years due to slow-moving factors such as ocean temperatures, both of which are known before the monsoon begins. Within a season, however, the timing and location of rainfall are very difficult to forecast, because the atmosphere is chaotic and small differences in initial conditions can produce very different rainfall patterns. Deviations from typical rainfall are therefore unlikely to be correlated with shocks to household outcomes realised after the onset of the monsoon.

Locations nonetheless differ systematically in their exposure to these shocks, that is, in the flooding they experience for a given realisation of $\mathbf{r}$. Random variation in rainfall therefore does not generate random variation in flooding.

Borusyak and Hull (2023) show that the bias from non-random exposure to exogenous shocks can be avoided by controlling for expected treatment. In our application, this corresponds to controlling for expected flooding, $\mathbb{E}[Flood_j]$, where the expectation is taken over all possible realisations of $\mathbf{r}$, given what is known before the monsoon begins.

We cannot compute this expectation directly, since we do not know the exact functional form of w_{jt} and in all likelihood observe only a subset of $\mathbf{X}$. Nor can we estimate it by averaging satellite-observed flooding across past monsoons, since $\mathbf{X}$ may change over time, as embankments are built and channels silt up.

We instead use a hydrological model, which predicts flooding in each hexagon on each day as a function of rainfall and the subset of characteristics that we observe, $\hat{\mathbf{X}}$,

$$H_{jt} = H_{jt}(\mathbf{r}_{1:t};\, \hat{\mathbf{X}}) \tag{5}$$

Following Borusyak and Hull (2026), we estimate the function $\hat{s}(\cdot)$ that combines these daily predictions into the best predictor of the satellite flood index, as described in Section 4.4,

$$\widetilde{Flood}_j(\mathbf{r}) = \hat{s}(H_{j1}, \ldots, H_{jT}) \tag{6}$$

For each hexagon, we then compute the deviation of predicted flooding under 2022 rainfall from its expectation across possible monsoons,

$$Z_j = \widetilde{Flood}_j(\mathbf{r}^{2022}) - \mathbb{E}\left[\widetilde{Flood}_j(\mathbf{r})\right] \tag{7}$$

We aggregate Z_j across hexagons to construct an instrument for each measure of flooding. The instrument for $F_i(k)$, $Z_i(k)$, is the mean of Z_j over the hexagons within k km of household i, and the instruments for $F_i^{\text{own}}(c)$ and $F_i^{\text{near}}(c)$, $Z_i^{\text{own}}(c)$ and $Z_i^{\text{near}}(c)$, are the means over the hexagons within c km and between c and 15 km.

Using these instruments, we estimate equations 1 and 2 by two-stage least squares. For equation 2, the two first-stage equations are

$$F_i^{\text{own}}(c) = \pi_1 Z_i^{\text{own}}(c) + \pi_2 Z_i^{\text{near}}(c) + \gamma_{D(i)} + u_i, \tag{8}$$

$$F_i^{\text{near}}(c) = \pi_3 Z_i^{\text{own}}(c) + \pi_4 Z_i^{\text{near}}(c) + \delta_{D(i)} + v_i \tag{9}$$

and the second stage is

$$Y_i = \beta_{\text{own}} \widehat{F}_i^{\text{own}}(c) + \beta_{\text{near}} \widehat{F}_i^{\text{near}}(c) + \alpha_{D(i)} + \varepsilon_i \tag{10}$$

where $\widehat{F}_i^{\text{own}}(c)$ and $\widehat{F}_i^{\text{near}}(c)$ are the fitted values from the first stage.

4.3 Identifying assumptions

Let Ω denote all information available before the monsoon begins, including $\mathbf{X}$ and slow-moving factors, such as ocean temperatures, that shift the distribution of rainfall. Two-stage least squares yields consistent estimates under two assumptions.

Assumption 1 (Correct rainfall distribution)**.** *The distribution over which we take the expectation in equation 7 is the true distribution of rainfall given* Ω*,*

$$\mathcal{R}(\cdot \mid \Omega) = \mathcal{R}_0(\cdot \mid \Omega)$$

Assumption 2 (Conditionally exogenous rainfall)**.** *Which monsoon occurred, out of those that were possible given* Ω*, is unrelated to unobserved determinants of household outcomes,*

$$\mathbb{E}[\varepsilon_i \mid \mathbf{r}, \Omega] = \mathbb{E}[\varepsilon_i \mid \Omega]$$

Assumption 2 does not require rainfall to be unconditionally randomly assigned across locations. To see this, decompose the error into a component determined before the monsoon and a component realised after its onset, $\varepsilon_i = \mathbb{E}[\varepsilon_i \mid \Omega] + \nu_i$. The assumption is then equivalent to $\mathbb{E}[\nu_i \mid \mathbf{r}, \Omega] = 0$. The predetermined component may be correlated with topography, infrastructure or expected flooding. What is required is only that deviations of realised rainfall from what was expected given Ω are unrelated to shocks to household outcomes realised after the onset of the monsoon.

Neither assumption requires the hydrological model to be correct. If its functional form is wrong or $\hat{\mathbf{X}}$ omits relevant characteristics, the instrument is still recentered over the true distribution of rainfall, so errors in the model weaken the first stage but do not bias the estimates. This would not hold if we instead controlled for the model's expected flooding, since the part of expected flooding that the model misses would remain in the error.

4.4 Implementation

We simulate flooding with LISFLOOD-FP (Bates et al., 2010; Neal et al., 2012), a hydrological model designed for large areas. The model takes daily rainfall and topography as inputs (Section 3) and outputs daily flood depth and extent in each hexagon. We input rainfall for July and August and run the model for five months, so that it captures the period over which floodwater drains. This yields 151 days of depth and extent, or 302 outputs per hexagon. We simulate flooding under 2022 rainfall and under the rainfall of each monsoon from 1998 to 2021. In doing so, we assume that these historical monsoons represent the distribution of monsoons that could have occurred in 2022. As an alternative, we use the 51 members of the ECMWF seasonal forecast ensemble initialised in May 2022, which condition on the state of the oceans and atmosphere before the monsoon.

To aggregate the hydrological outputs into a prediction of the satellite flood index, we could apply the formula we use to construct the index itself, the first principal component of the monthly averages (Section 3). The model's dynamics may differ from those of real floods, however, so we instead treat the 302 outputs as candidate instruments and follow the optimal instrument approach of Borusyak and Hull (2026), which builds on Chamberlain (1987). We estimate $\hat{s}(\cdot)$ as the best predictor of the satellite index given the 302 outputs, using a random forest. Intuitively, the random forest

selects the weight on each day, and on depth relative to extent, that best predicts the satellite index. To avoid over-fitting, we cross-fit (Chernozhukov et al., 2018). We assign hexagons to five folds, clustered by H3 resolution-6 hexagon of approximately 36 km^2, to account for spatially correlated errors, and predict each fold with a forest trained on the other four.

Since $\hat{s}(\cdot)$ is a fixed function of the model's outputs, we apply it to the outputs from each historical monsoon to compute $\widetilde{Flood}_j(\mathbf{r})$ and hence the expectation in equation 7.

4.5 Inference

We report Conley (1999) standard errors, which allow errors to be spatially correlated up to a distance cutoff. Since our instrument is generated by a single rainfall realisation propagating through the hydrological network, we set the cutoff at the scale over which the instrument is spatially correlated. We estimate this scale from the instrument's correlogram, which plots the correlation between each hexagon's instrument and those of its neighbours at increasing distances, and set the cutoff where the correlation falls to zero. The cutoff varies with the sample and the measure of flooding, from about 10 km for flooding within a small radius to about 30 km for the own-versus-nearby regressions.

Since all households share one realisation of rainfall, we also conduct randomisation inference. We construct placebo instruments by treating each historical monsoon as if it were the realised one, and compare our estimates with the distribution of estimates under these placebos.

4.6 First-stage results

Figure 1 summarises the first stage for each specification we estimate. Panel A plots the first-stage F-statistic for equation 1 at each radius k. Panel B plots the Kleibergen–Paap statistic for equation 2 at each cutoff c. The instruments are strong across all specifications, with statistics ranging from 10 to 147. Panel C plots the correlation between $Z_i^{\text{own}}(c)$ and $Z_i^{\text{near}}(c)$ within districts at each cutoff c, which determines how much independent variation we have to identify β_{own} and β_{near}.

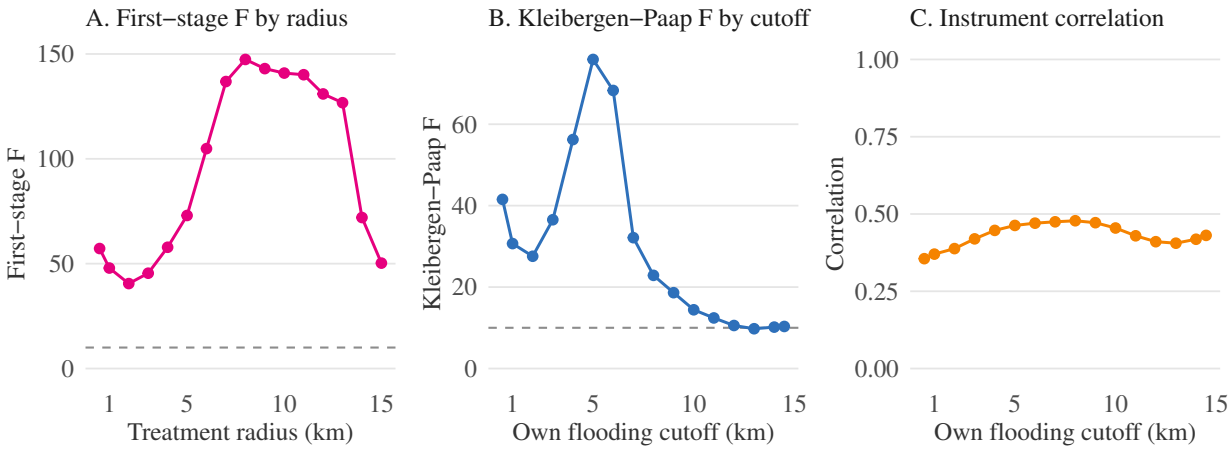


Figure 1: First stage, by treatment radius and by own cutoff

Notes: Panel A plots the F-statistic from a regression of flooding within each radius on its instrument, the mean recentered hydrological prediction over the same radius. Panel B plots the Kleibergen-Paap F-statistic for the specification with two regressors, flooding within the cutoff and flooding from the cutoff to 15 km, each instrumented by the recentered prediction over the same area. Panel C plots the correlation between the two instruments in Panel B, the recentered prediction within the cutoff and from the cutoff to 15 km, after removing district means. The dashed line marks F = 10. All regressions include district fixed effects and use Conley standard errors with a cutoff set from the instrument's spatial correlogram. The sample is the 4,536 households surveyed in both waves for whom we observe a food price index.

5 EMPIRICAL RESULTS

5.1 Household consumption depends more on flooding in the surrounding area than on flooding at home

We begin with our key result, that a household's consumption loss depends more on flooding in the surrounding area than on flooding at its own location. We measure consumption as real expenditure, that is, household expenditure in the previous month, adjusted for household composition and deflated by a price index constructed from the food prices that households report paying (Section 3).

Panel A of Figure 2 reports estimates of equation 1 for radii k from 0.5 to 15 km. We scale each coefficient by the difference between the 10th and 90th percentiles of the within-district distribution of $F_i(k)$, so that it gives the effect of moving a household from a lightly to a heavily flooded area, at that radius. When the radius is just 1 km around a home, real expenditure is 6.8% lower for highly exposed households ($p = 0.05$). Widening the radius raises the estimated loss to 9.4% at 10 km ($p = 0.02$) and 10.1% at 15 km ($p = 0.01$).

In Panel B we then decompose the total effect of flooding within 15 km into the contribution of nearer and more distant flooding, by including separate regressors for flooding within a cutoff

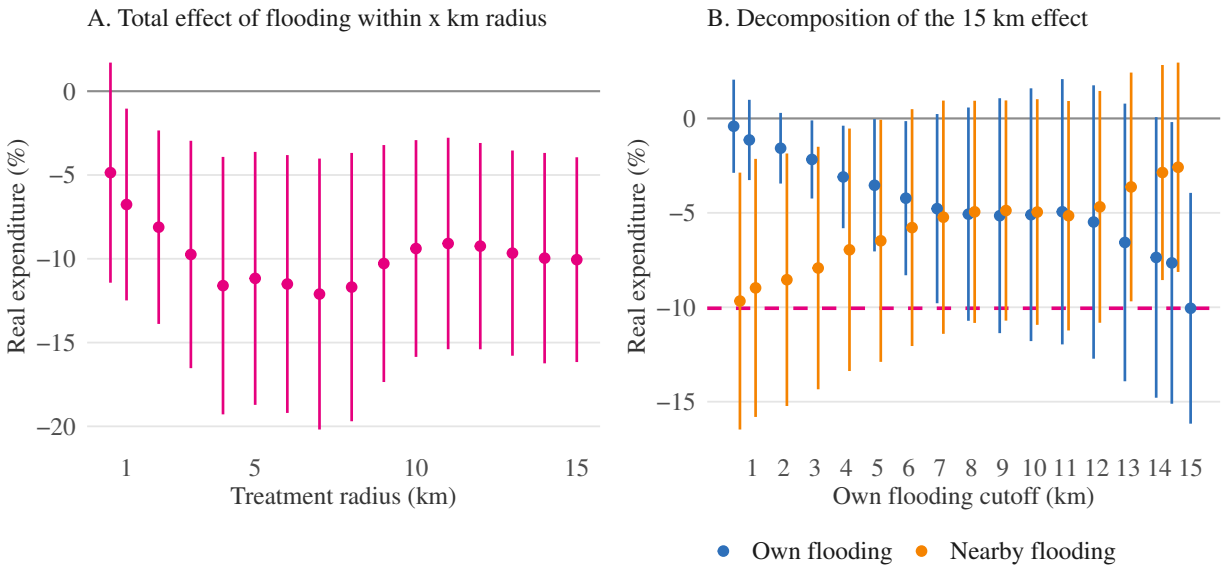


Figure 2: Effect of flooding on real expenditure, by treatment radius and by own cutoff

Notes: Panel A plots the two-stage least squares coefficient on flooding within each radius, instrumented by the mean recentered hydrological prediction over the same radius, multiplied by the within-district difference between the 10th and 90th percentiles of that regressor. Panel B plots, for each cutoff, the coefficients on flooding within the cutoff (blue) and flooding from the cutoff to 15 km (orange) from one regression with both regressors, each instrumented by the recentered prediction over the same area. Each coefficient in Panel B is multiplied by the average change in its regressor when flooding within 15 km moves from its 10th to its 90th percentile within districts. The dashed line marks the 15 km estimate from Panel A, which is also the point at 15 km. The outcome is log monthly household expenditure per adult equivalent, deflated by a household food price index, so estimates are in percent. Whiskers are 90% confidence intervals from Conley standard errors with a Bartlett kernel and a cutoff set from the instrument's spatial correlogram. All regressions include district fixed effects. The sample is the 4,536 households surveyed in both waves for whom we observe a food price index.

distance of c km and flooding beyond that cutoff, up to 15 km. Moving a household from the 10th to the 90th percentile of flooding within 15 km raises both its own flooding, within c km, and its nearby flooding, between c and 15 km. We compute the average change in each component that accompanies this move, and multiply each by its coefficient from equation 2. The two contributions sum approximately to the total effect within 15 km from Panel A, shown by the dashed line.

This exercise shows that flooding within 1 km of home accounts for only about 11% of the expenditure decline, while flooding more than 5 km away accounts for 64%, and flooding more than 10 km away for 49%. The contribution of own flooding is small and statistically insignificant at narrow cutoffs, and rises only as the cutoff widens to include flooding several kilometres away.

5.2 Spillovers through labour, product and credit markets

Why does flooding several kilometres from a household affect its consumption? We find that the answer lies in spatial linkages through labour, product and credit markets.

Suppose that the area within 1 km of each household is a self-contained village existing in autarky. Consumption would depend only on output in the local area, and flooding elsewhere would be irrelevant. If villages are instead connected through goods, labour and credit markets, a flood in one village can affect people living in others. A household may lose work in a flooded village, be unable to reach its usual market or need to support flooded relatives. More generally, a flood in one location can reduce the demand for labour or the supply of food in a local market, and so change wages and prices in many other locations. The costs of flooding in a household's own village are then shared with people elsewhere, providing partial insurance against local shocks, and the household in turn shares in the costs of flooding in other villages.

In the limit in which everything that a household consumes is traded and there is no cost of commuting or transporting goods, only flooding across the economy as a whole would matter. When trade and commuting are costly, and these costs rise with distance, spillovers are concentrated in space, so flooding in the area around a household's home becomes an important determinant of its outcomes. In the model of Section 6, consumption loss is determined by the flood-induced change in market access, which is effectively a distance-weighted average of flooding in the surrounding area.

If spillovers operate through markets, they should affect the quantities and prices determined in those markets. Spillovers through labour markets should affect employment, wages and commuting, those through goods markets should affect prices, and those through credit markets should affect borrowing and asset sales. Impacts on these outcomes are shown in Figures 3 and 4.

The top row of Figure 3 shows impacts on commuting, measured by the likelihood that a household member commutes more than 1 km to work. A 10th-to-90th percentile increase in flooding within 1 km of a house raises by 6 percentage points the likelihood that a household member commutes more than 1 km to work. This suggests that households respond to local flooding by seeking work further afield. But the effect shrinks as the exposure radius widens, reaching roughly zero at 15 km. The distance decomposition helps explain why. Holding flooding in other distance bands constant, flooding within 5 km raises commuting by about 4 percentage points. By contrast, flooding 5–15 km away lowers it enough to offset that increase. This suggests that households flooded at home but surrounded by dry land can seek work elsewhere, but that this becomes increasingly

difficult as more of the surrounding area is flooded. These results are in line with Brooks and Donovan (2020), who find that bridges, by maintaining access to outside labour markets, protect earnings from local flooding.

Wage employment shows a similar pattern (middle row), measured as the share of working-age men in the household who work for others. Flooding within 1 km reduces wage employment by 1.5 percentage points ($p = 0.60$), rising to 3.4 percentage points within 5 km and 8.9 percentage points within 15 km ($p = 0.01$). The effect is driven by households whose male members all worked in agriculture before the floods, for whom flooding within 15 km reduces wage employment by 16.9 percentage points, compared with 0.9 percentage points for other households (Appendix B.1).

Panel B shows that the effect is due almost entirely to nearby flooding. With a 1 km cutoff, nearby flooding accounts for a fall of 9.3 percentage points, whereas own flooding slightly increases wage employment, by 0.5 percentage points ($p = 0.26$). The pattern persists as the cutoff widens. With a 5 km cutoff, nearby flooding accounts for a fall of 11.0 percentage points and own flooding for a rise of 2.2 ($p = 0.16$). This pattern is consistent with households supplying more labour in response to their own losses, while flooding in the surrounding area reduces the demand for their labour.

Part of the fall in wage employment is offset by a rise in self-employment (Appendix Figure 14). Moving a household from the 10th to the 90th percentile of flooding within 15 km raises the share of its men in self-employment by 4.9 percentage points ($p = 0.01$), and total employment falls by only 3.2 percentage points ($p = 0.10$). As with wage employment, the rise in self-employment is driven by nearby flooding, which accounts for 5.0 percentage points with a 1 km cutoff. This is consistent with labour rationing, whereby workers displaced from wage work turn to lower-return forms of self-employment (Breza et al., 2021). We find no effect of flooding on daily wages. This is in line with wages being rigid downwards in village labour markets (Kaur, 2019), so that a fall in the demand for labour reduces employment rather than wages.

The bottom row of Figure 3 shows the effect of flooding on food prices, measured using a household price index constructed from the prices that households report paying for food (Section 3). Panel A shows that flooding raises food prices, and that the effect grows with the radius over which we measure flooding. Flooding within 1 km raises food prices by 4.4% ($p = 0.07$), rising to 6.2% within 5 km and 8.1% within 15 km ($p = 0.02$). Panel B shows that this effect is due almost entirely to nearby flooding. With a 1 km cutoff, nearby flooding raises food prices by 7.6%, whereas own flooding has no effect (0.6%, $p = 0.25$). Even at 10 km, flooding beyond the cutoff accounts for the entire effect. This is consistent with food prices being set in markets that serve a wider area

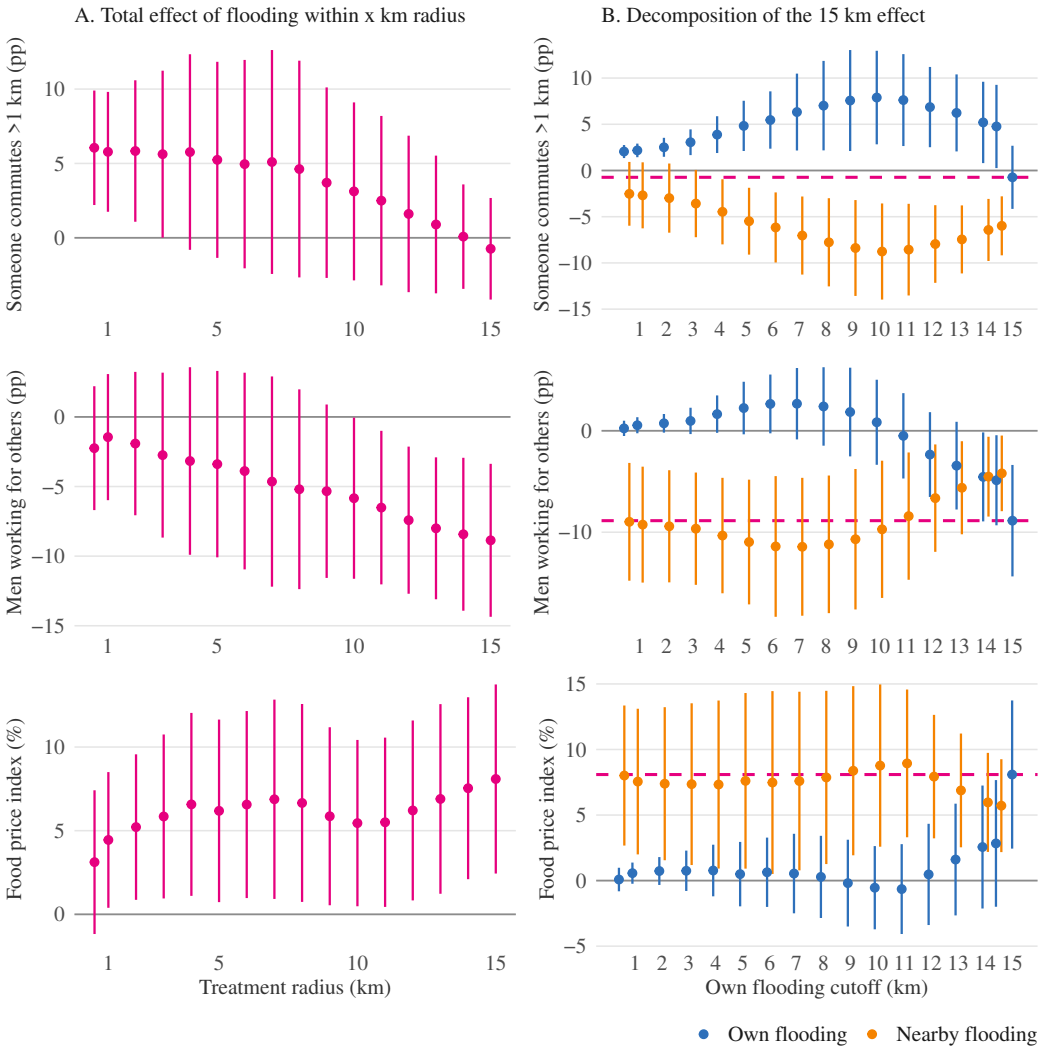


Figure 3: Effect of flooding on commuting, wage employment and the food price index

Notes: Panel A plots the two-stage least squares coefficient on flooding within each radius, instrumented by the mean recentered hydrological prediction over the same radius, multiplied by the within-district difference between the 10th and 90th percentiles of that regressor. Panel B plots, for each cutoff, the coefficients on flooding within the cutoff (blue) and flooding from the cutoff to 15 km (orange) from one regression with both regressors, each instrumented by the recentered prediction over the same area. Each coefficient in Panel B is multiplied by the average change in its regressor when flooding within 15 km moves from its 10th to its 90th percentile within districts. The dashed line marks the 15 km estimate from Panel A, which is also the point at 15 km. The outcome in the top row is an indicator for any of the household's men aged 18 to 65 working more than 1 km from home, controlling for the number of such men. The outcome in the middle row is the share of the household's men aged 18 to 65 who work for others, controlling for the number of such men. The outcome in the bottom row is the log of a household food price index, the value of the food the household consumed divided by the same quantities valued at sample-average unit prices, so estimates are in percent. In all panels the unit of observation is a household surveyed in both waves. In the top and middle rows the sample is households with at least one man aged 18 to 65 whose pre-flood sector we observe (N = 4,478). In the bottom row the sample is households for whom we observe a food price index (N = 4,536). Whiskers are 90% confidence intervals from Conley standard errors with a Bartlett kernel and a cutoff set from the instrument's spatial correlogram. All regressions include district fixed effects.

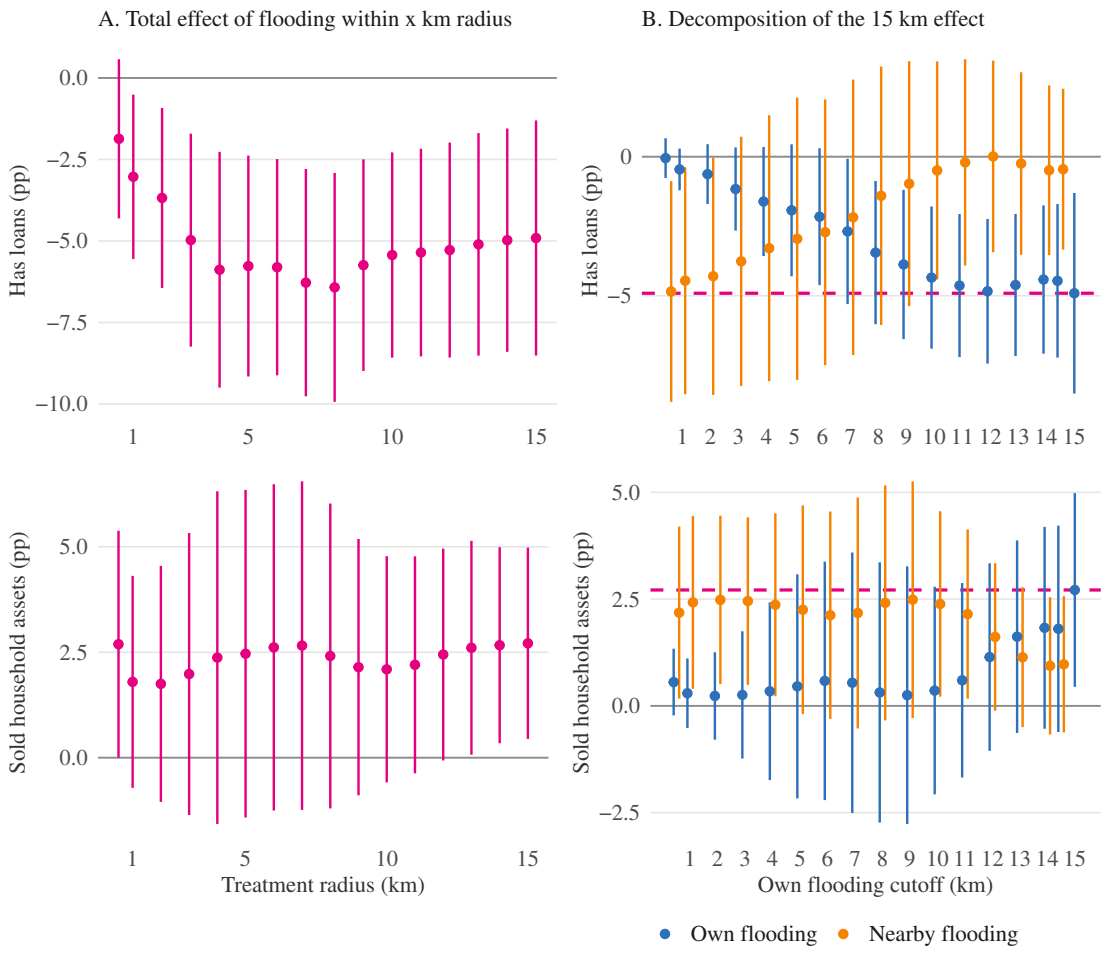


Figure 4: Effect of flooding on loans and asset sales, by treatment radius and by own cutoff

Notes: Panel A plots the two-stage least squares coefficient on flooding within each radius, instrumented by the mean recentered hydrological prediction over the same radius, multiplied by the within-district difference between the 10th and 90th percentiles of that regressor. Panel B plots, for each cutoff, the coefficients on flooding within the cutoff (blue) and flooding from the cutoff to 15 km (orange) from one regression with both regressors, each instrumented by the recentered prediction over the same area. Each coefficient in Panel B is multiplied by the average change in its regressor when flooding within 15 km moves from its 10th to its 90th percentile within districts. The dashed line marks the 15 km estimate from Panel A, which is also the point at 15 km. The outcome in the top row is an indicator for the household having loans outstanding at the time of the 2023 survey. The outcome in the bottom row is an indicator for the household having sold household assets or goods since the 2022 monsoon. In all panels the unit of observation is a household surveyed in both waves, and the sample is all such households (N = 4,683). Whiskers are 90% confidence intervals from Conley standard errors with a Bartlett kernel and a cutoff set from the instrument's spatial correlogram. All regressions include district fixed effects.

than a single village, so that what matters for the price a household pays is the loss of supply across that area, rather than in its own village. This is in line with Allen and Atkin (2022), who show that highway expansion in India made local prices less responsive to local yields and more responsive to yields elsewhere.

Figure 4 shows the effect of flooding on borrowing and asset sales, measured by whether the household has taken a loan and whether it has sold household assets since the floods. Panel A shows that flooding reduces borrowing. Flooding within 1 km reduces the share of households with loans by 3.0 percentage points ($p = 0.05$), 5.8 within 5 km and 4.9 within 15 km ($p = 0.03$). Panel B shows that, at narrow cutoffs, this effect is due almost entirely to nearby flooding. With a 1 km cutoff, nearby flooding reduces borrowing by 4.5 percentage points, whereas own flooding has no effect (0.5, $p = 0.32$). Flooding also raises asset sales, by 2.7 percentage points within 15 km ($p = 0.05$), again almost entirely because of nearby flooding, which accounts for 2.5 percentage points with a 1 km cutoff and 2.5 with a 10 km cutoff. This suggests that households flooded at home can continue to borrow from lenders in the surrounding area, but that when the surrounding area is also flooded, credit becomes scarce and households instead sell assets. The rise in asset sales offsets only part of the fall in borrowing, possibly because assets are also harder to sell when many nearby households are affected.

5.3 Interpretation of magnitudes

5.3.1 Per-hexagon effects

To interpret the magnitudes of these results, we can decompose the total effect of flooding within 15 km into the effect of flooding in each hexagon at a given distance and the number of hexagons at that distance. Doing so, we show that the further a hexagon is from a household's home, the less its flooding matters for that household's outcomes. But there are many more hexagons the wider we draw the circle, so collectively the flooding in that area can matter a great deal.

Whilst we are not powered to run regressions with separate variables for flooding in each 1 km bin around the household, we can back out from the own and nearby coefficients in Panel B of Figure 2 the effect of one unit of the flood index in a single hexagon at a given distance. For example, to compute the effect of a hexagon 2 to 3 km away, we take the coefficient on nearby flooding when the cutoff is 2 km, subtract the coefficient when the cutoff is 3 km, and divide by the 144 hexagons of 170 m radius (resolution 9 of the H3 grid) that lie between 2 and 3 km away from a household. Intuitively, the two coefficients differ only in whether they include the hexagons from 2 to 3 km, so the difference reflects those hexagons' impact on consumption. We can also compute the effect of a hexagon 2 to 3 km away from the own coefficients, by taking the coefficient on own flooding when the cutoff is 3 km, subtracting the coefficient when the cutoff is 2 km, and dividing by the same 144 hexagons. Panel A of Figure 5 plots both versions.

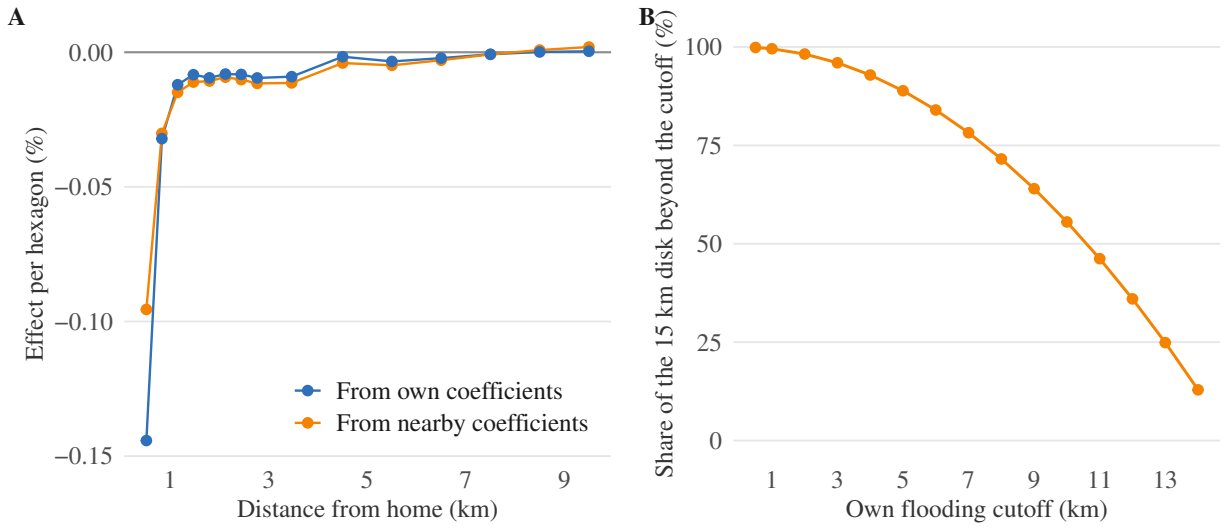


Figure 5: Area beyond the cutoff and effect of flooding per hexagon

Notes: Panel A plots the effect of one unit of the flood index in one hexagon on the outcome, by the hexagon's distance from the household. Each point is the difference between the coefficients from two regressions whose cutoffs sit at either end of the distance interval, divided by the mean number of hexagons of 170 m radius between the two cutoffs around the sample households, and plotted at the midpoint of the interval. Each regression has two regressors, flooding within the cutoff and flooding from the cutoff to 15 km, each instrumented by the mean recentered hydrological prediction over the same area. The orange series differences the coefficients on flooding from the cutoff to 15 km and the blue series the coefficients on flooding within the cutoff. The outcome is log monthly household expenditure per adult equivalent, deflated by a household food price index, so effects are in percent. The points carry no confidence intervals. All regressions include district fixed effects. The sample is the 4,536 households surveyed in both waves for whom we observe a food price index. Panel B plots the share of the area within 15 km of a household that lies beyond a circle of the given radius, computed from plain circle areas.

Panel A shows that a one-unit increase in the flood index in a hexagon half a kilometre from home lowers real expenditure by 0.10 to 0.14%. For a hexagon just under 1 km away the effect is 0.03%, and from 1 to 4 km it is about 0.01%. From 4 to 7 km it is a third of that. Panel B shows why these small effects add up. Of the area within 15 km of a household, 99% lies beyond 1 km and 89% beyond 5 km, so there are 37 hexagons within 1 km of a household and more than 6,000 between 1 and 15 km. Putting that together, increasing flood intensity for all hexagons within 1 km by one unit would lower real expenditure by 3.8% (37 hexagons times 0.10% each), whereas applying the same increase in intensity to all hexagons from 1 to 15 km would lower it by 21% (6,386 hexagons times 0.003% each).

5.3.2 Autarky benchmark

As a second exercise to help understand the magnitude of these effects, we can compare the direct impacts to an extremely rough autarky benchmark. We estimate that own flooding at one unit of

the flood index lowers agricultural output in the household's own hexagon by 33%. Since food comprises 56% of consumption for poor rural households in Sindh, this implies that a household in autarky, consuming only what its own location produces, would see its consumption fall by 18.5% in the season of the flood. Our consumption results, by contrast, find that one year after the floods, consumption in flooded locations is 3.8% lower per unit of the flood index, controlling for flooding in the surrounding area. Taking the linear functional form literally, this implies that for a location where only they were flooded, consumption falls by 3.8%, compared to the 18.5% autarky benchmark. The difference between these numbers reflects smoothing over time, since our consumption impacts are measured one year after the flood and the production estimates are contemporaneous, and smoothing over space.

5.4 Robustness

This section addresses potential concerns which could be driving our results. We first show that they are not driven by error in measuring household locations, by using objective housing damage to run a placebo test. We also show evidence which suggests that the allocation of aid, which favours directly exposed households, leads us to somewhat underestimate the direct impacts of flooding, but does not qualitatively change our conclusion that households can suffer large losses due to flooding in nearby areas.

5.4.1 Measurement error

Two types of measurement error could affect our estimates: error in measured flooding at a location, $Flood_j$, and in the recorded location of each household. Our IV approach already addresses the first, since the instrument is an independent measure of flooding, but may not address the second. Suppose we record household i at $g_i = \ell_i + \xi_i$, where ℓ_i is its true location. Its measured own flooding, $F_i^{\text{own}}(c)$, then contains an error approximately equal to the displacement ξ_i times the spatial gradient of flooding at ℓ_i, which is largest at the edges of flooded areas. Since we construct the instrument at the same recorded location, this error is correlated with the instrument. Nearby flooding then partly proxies for true own flooding, biasing β_{own} down and β_{near} up (Huber, 2023).

To test for this, we use physical damage as a placebo outcome, since flooding several kilometres away cannot damage a house. If location error inflated the effect of nearby flooding, nearby flooding would appear to cause damage. Figure 6 shows that it does not. With a 1 km cutoff, own flooding raises the share of buildings damaged by 5.9 percentage points and lowers crop greenness by 33%,

whereas nearby flooding has small and insignificant effects on both. Since location error biases own and nearby effects together, this also suggests that we do not substantially underestimate the effect of own flooding. The consumption effects of nearby flooding are therefore unlikely to be an artefact of location error.

Our estimand is nonetheless broader than the effect of flooding at a household's home. Moving a household from the 10th to the 90th percentile of flooding within 1 km raises the share of buildings damaged by 5.9 percentage points. Scaling our own-flooding estimates by this change would give an illustrative effect of home damage on consumption, although flooding also affects consumption through channels other than housing damage.

5.4.2 Aid

A second concern is that our estimates are net of aid. If relief targeted physically exposed households, it could offset the effect of own flooding, and so explain why flooding close to the home has little effect on consumption. We do not have random variation in aid. We can, however, estimate the effect of flooding separately for the two-thirds of households who report receiving no relief.

Figure 7 repeats Figure 2 for households who report receiving no relief. For these households, the effect of flooding is larger at every radius, reaching 15% within 15 km. Own flooding also accounts for a larger share of the effect. Panel C compares the share of the 15 km effect due to own flooding for all households and for households without relief. With a 1 km cutoff, this share rises from 11% to 24%, and it is higher for households without relief at every cutoff. For households who received relief, the effect of flooding is smaller and is driven almost entirely by nearby (not own) flooding (Appendix Figure 13).

Together, these results suggest that households with more own flooding received more relief, which helped them maintain consumption, whereas relief did not flow to households on the basis of flooding in the surrounding area. Our estimates for the full sample are therefore likely to understate the effect of own flooding, and so to overstate the share of the effect due to nearby flooding. Aid does not, however, explain away the effect of nearby flooding. Even among households who received no relief, flooding more than 1 km away accounts for 76% of the effect, and flooding more than 5 km away for 58%.

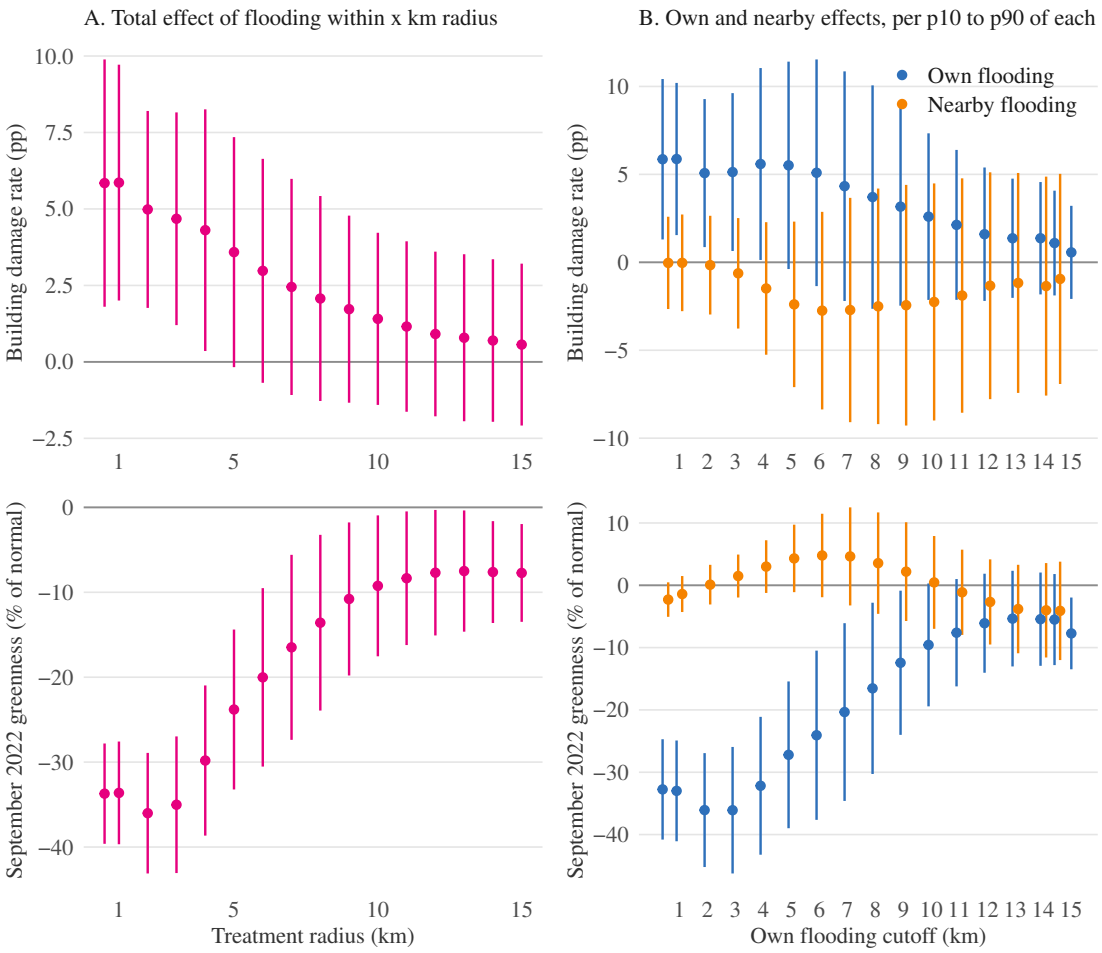


Figure 6: Effect of flooding on housing damage and crop greenness, by treatment radius and by own cutoff

Notes: Panel A plots the two-stage least squares coefficient on flooding within each radius, instrumented by the mean recentered hydrological prediction over the same radius. Panel B plots, for each cutoff, the coefficients on flooding within the cutoff (blue) and flooding from the cutoff to 15 km (orange) from one regression with both regressors, each instrumented by the recentered prediction over the same area. Every coefficient is multiplied by the within-district difference between the 10th and 90th percentiles of its own regressor. The outcome in the top row is the number of a hexagon's buildings recorded as damaged in the Government of Sindh's damage census, divided by the number of buildings in the VIDA footprint layer. The outcome in the bottom row is mean September 2022 NDVI of cropland in the hexagon, controlling for its 2016 to 2020 September mean, with coefficients divided by the sample mean of that history (0.53) so they read as percent of normal greenness. In all panels the unit of observation is a hexagon of 460 m radius in a union council where we surveyed households. In the top row the sample is hexagons with at least 50 buildings, weighted by building count and controlling for log buildings (N = 2,498). In the bottom row the sample is hexagons containing cropland in the ESA WorldCover map, with NDVI averaged over their cropland cells (N = 4,075). Whiskers are 90% confidence intervals from Conley standard errors with a Bartlett kernel and a cutoff set from the instrument's spatial correlogram. All regressions include district fixed effects.

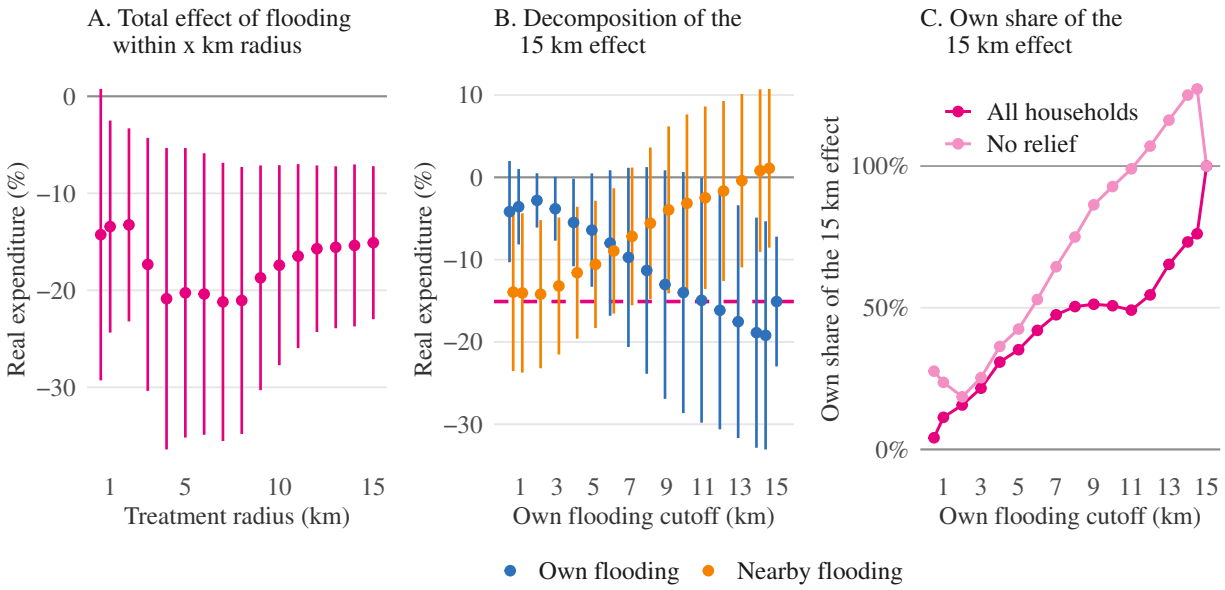


Figure 7: Effect of flooding on real expenditure, households reporting no relief

Notes: All estimates are the effect of flooding for households reporting no flood relief, from one regression per point on all households that includes flooding, an indicator for reporting relief and their interaction, instrumented by the recentered hydrological prediction and its interaction. Panel A plots this coefficient for flooding within each radius, multiplied by the within-district difference between the 10th and 90th percentiles of that regressor. Panel B plots, for each cutoff, the coefficients on flooding within the cutoff (blue) and flooding from the cutoff to 15 km (orange) from one regression with both regressors. Each coefficient in Panel B is multiplied by the average change in its regressor when flooding within 15 km moves from its 10th to its 90th percentile within districts. The dashed line marks the 15 km estimate from Panel A, which is also the point at 15 km. Panel C plots the own flooding point in Panel B divided by the 15 km estimate from Panel A, for households reporting no relief (pale pink) and for all households from regressions without the relief interaction (pink). The outcome is log monthly household expenditure per adult equivalent, deflated by a household food price index, so estimates are in percent. The unit of observation is a household surveyed in both waves for whom we observe a food price index and whether they received flood relief (N = 4,178). Whiskers are 90% confidence intervals from Conley standard errors with a Bartlett kernel and a cutoff set from the instrument's spatial correlogram. All regressions include district fixed effects.

5.5 Interpreting relative effects

The results above show that a household's consumption loss depends on flooding in the surrounding area in addition to flooding at its own location. In specifying our regressions, we remained agnostic about the channels through which spillovers operate, imposing only that their effects depend on the distance between locations. This was enough to test the assumption, implicit in much disaster aid policy, that physical exposure is the main determinant of loss. To know whether these estimates can guide the targeting of aid on their own, however, we need to understand how the predicted values from our regressions relate to the total loss in each location caused by the entire flood. Appendix A sets out a potential outcomes framework that relates the two, adapted from Hudgens and Halloran (2008), which we summarise here.

Let $Y_i(\mathbf{f})$ denote an outcome, say consumption, in location i as a function of the whole vector of flooding, $\mathbf{f} = (f_i, f_{-i})$, where f_{-i} is flooding in all locations except i. The direct effect of flooding on i, $\mathrm{DE}_i \equiv Y_i(f_i, f_{-i}) - Y_i(0, f_{-i})$, is the change in its consumption caused by its own flooding, holding fixed flooding everywhere else. The indirect effect, $\mathrm{IE}_i \equiv Y_i(0, f_{-i}) - Y_i(0, \mathbf{0})$, is the impact on consumption in i of flooding in all locations except i. The total effect of the flood on i, $\mathrm{TE}_i \equiv Y_i(\mathbf{f}) - Y_i(\mathbf{0})$, is the sum of the direct and indirect effects.

Many studies of disasters compare affected with unaffected locations. If flooding is randomly assigned, this comparison equals the expected direct effect on flooded locations, plus the difference in expected spillovers between flooded and unflooded locations, which we call the net spillover. What the comparison recovers therefore depends on the net spillover. If there are no spillovers, for example because the flood affects an infinitesimally small area, the comparison recovers the direct effect, which then equals the total effect. If there are spillovers, but they are the same at every location, as in Nakamura and Steinsson (2014), Chodorow-Reich (2019, 2020) and Wolf (2023), the comparison recovers the direct effect but not the total effect, since the common spillover is differenced out. Adão et al. (2026) show that this holds only when locations respond symmetrically to shocks and there are no spatial frictions. With spatial frictions, spillovers are heterogeneous in space, and the comparison can be larger or smaller than either the direct or the total effect, depending on the net spillover. If spillovers are larger from nearby locations, however, then since flooding is spatially correlated, flooded locations are likely to suffer larger spillovers than unflooded ones. If flooding lowers consumption, the comparison then most likely lies between the direct effect and the total effect on flooded locations.

Our regressions do not compare flooded with unflooded locations. Instead, they compare locations within the same district that differ in mean flooding within c km of them, and from c to 15 km. Since these regressors average flooding over many locations, it helps to take a linear approximation of $Y_i(\mathbf{f})$ to interpret their coefficients. To first order, $Y_i(\mathbf{f}) = Y_i(\mathbf{0}) + \sum_j b_{ij} f_j$, where f_j is the flood index in location j and b_{ij} is the marginal effect of flooding in j on consumption in i, so that the total effect is $\mathrm{TE}_i = \sum_j b_{ij} f_j$. Splitting this into terms by distance from i, we can compare it to our regression specification from Section 4,

$$Y_i = \sum_{j:\, d_{ij} \leq c} b_{ij} f_j \quad + \sum_{j:\, c < d_{ij} \leq 15} b_{ij} f_j + \sum_{j:\, d_{ij} > 15} b_{ij} f_j + Y_i(\mathbf{0})$$

$$Y_i = \beta_{\text{own}} F_i^{\text{own}}(c) + \beta_{\text{near}} F_i^{\text{near}}(c) \; + \alpha_{D(i)} + \varepsilon_i$$

where d_{ij} is the straight-line distance in km between i and j. We now ask how the predicted loss in each location, $\hat{\beta}_{\text{own}} F_i^{\text{own}}(c) + \hat{\beta}_{\text{near}} F_i^{\text{near}}(c)$, relates to the total loss, TE_i.

Suppose that flooding more than 15 km away does not affect i, and that flooding in every location within a distance band has the same effect on i. The predicted loss then equals the total effect of the flood on i. These assumptions are, however, highly restrictive. We relax them in turn.

If flooding more than 15 km away does affect i, but this effect is constant across locations within a district, then it will be absorbed by the district fixed effects, leaving the estimated coefficients unchanged. The predicted loss now leaves out this common spillover, and so understates the total effect of the flood on each location if the spillover lowers consumption. Since this missing component does not vary within a district, the resulting predictions will correctly rank locations by total loss within a district, but without knowing this district-level spillover we cannot compare across districts.

If instead the spillover from beyond 15 km varies across locations within a district, and is correlated with flooding within 15 km, as is likely since flooding is spatially correlated, part of it will be picked up by $\hat{\beta}_{\text{own}}$ and $\hat{\beta}_{\text{near}}$. From the perspective of ranking locations by their loss, this bias is not a problem, and may even be useful, since it means that our estimates capture the part of the impact of flooding beyond 15 km that can be predicted by flooding within 15 km. The predicted loss for each location still misses spillover effects that cannot be predicted by flooding within 15 km, or that are common to all locations in a district.

Finally, locations within c km of i, or between c and 15 km, may differ in their effect on i, for example because closer locations matter more, and our power limits how flexibly we can let these effects vary with distance. Each of our coefficients is then a weighted average of the effects of the locations it covers, with weights determined by the typical spatial pattern of flooding.

Viewed through this lens, our reduced-form estimates are useful for ranking locations by loss within a district, but not across districts. These problems apply equally to our IV estimates, since our IV strategy addresses only non-random flooding. To estimate the total loss in each location, we therefore need to impose more structure. Taking a stance on the channels through which spillovers operate, as our model of Section 6 does, allows us to capture spillovers from beyond 15 km, including how they vary across space and across districts. It also allows us to predict the loss in each location as a function of flooding in all locations, rather than only based on mean flooding within 15 km.

6 A SPATIAL MODEL OF DISASTER INCIDENCE

Our reduced-form results show that a household's losses depend on flooding both close to its home and in the surrounding area, so the incidence of a flood, meaning the distribution of welfare losses across locations, need not coincide with its physical footprint. As we have just argued, however, they cannot recover the total loss in each location, since they capture only flooding within 15 km and district fixed effects absorb any effects common to all households in a district.

We therefore build a spatial model in which many locations are linked through costly trade, following Eaton and Kortum (2002), and costly commuting, following the quantitative spatial literature (Ahlfeldt et al., 2015). The costs of a flood are shared through goods and labour markets, the two main channels through which we find spillovers in the data. Taking a stand on how shocks propagate allows us to recover the welfare loss in every location and to evaluate counterfactual floods and aid policies.

6.1 Setup

We model a closed economy with N locations, inhabited by workers, landowners and firms. Each location i has a fixed residential population of L_i workers, who supply labour in the location and sector of their choice. Workers derive utility from consuming goods and housing services, the latter from a non-tradable endowment of housing. Each location also has a single landowner who owns the land and consumes goods bought with land rents.

Goods market: The goods market builds on Eaton and Kortum (2002), with a few departures. There are two sectors, agriculture A and non-agriculture M, each with a continuum of goods which can be produced anywhere. Agricultural goods are produced from labour and land using Cobb–Douglas technology with labour share α; non-agricultural goods are produced from labour alone. Locations differ in their average productivity and their idiosyncratic productivity for each good. Productivity for good u in sector s in location j is $z_j^s(u)$ – drawn from a Fréchet distribution with scale parameter T_j^s and shape parameter θ_T. T_j^s captures the absolute advantage of a location in that sector, whilst θ_T governs dispersion across goods. A location can be a strong producer of food and a weak producer of everything else. The unit cost of producing good u in sector s in location j is:

$$\frac{c_j^s}{z_j^s(u)} \quad \text{with} \quad c_j^A \propto (w_j^A)^\alpha r_j^{1-\alpha}, \quad c_j^M = w_j^M \tag{11}$$

where w_j^s is the sector-s wage rate and r_j is the land rental rate. Firms operate in perfect competition, so this will also be the mill price of good u in location j.

Workers and landowners split their spending between the two sectors in Cobb–Douglas shares, with agricultural share χ, and have identical constant-elasticity-of-substitution (CES) preferences over the continuum of goods within each sector, with elasticity of substitution σ. They can buy goods from anywhere, but shipping is subject to iceberg trade costs $\kappa_{ij}^G = t_{ij}^{\tau^G}$, rising in the travel time t_{ij} between the two locations, so that κ_{ij}^G units must be shipped from j for one to arrive in i. They source each good from the least-cost location, and the resulting share of i's sector-s spending that goes to j is:

$$\pi_{ij}^{G,s} = \frac{T_j^s \left(c_j^s \kappa_{ij}^G\right)^{-\theta_T}}{\sum_k T_k^s \left(c_k^s \kappa_{ik}^G\right)^{-\theta_T}} \tag{12}$$

All else equal, i will buy more from j if j has higher productivity, lower wages and land rents, or is closer. Trade costs and the elasticity are common to both sectors, but the two trade networks differ because productivities and bundle costs do. The sector price index in i falls with its goods-market access in that sector: $P_i^s \propto (\mathrm{GMA}_i^s)^{-1}$, where $\mathrm{GMA}_i^s \equiv \left[\sum_k T_k^s (c_k^s \kappa_{ik}^G)^{-\theta_T}\right]^{1/\theta_T}$. The cost of living combines the two, $P_i = (P_i^A)^{\chi}(P_i^M)^{1-\chi}$.

Labour market: Workers reside in i but choose where to work and in which sector. All else equal, workers prefer jobs that pay higher wages w_j^s and have lower commuting costs $\tau_{ij} = \exp(\tau^L t_{ij})$. They also draw an idiosyncratic taste for each location-sector pair ζ_{ij}^s from a Fréchet distribution with shape parameter θ_L. Putting this together, the share of i's residents in job (j, s) is

$$\pi_{ij}^s = \frac{(w_j^s/\tau_{ij})^{\theta_L}}{\sum_k \sum_t (w_k^t/\tau_{ik})^{\theta_L}} \tag{13}$$

rising in the wage on offer and falling in distance, with θ_L governing how much workers sort on the financial benefits of jobs rather than idiosyncratic preferences. Both sectors at j face the same commuting cost, so distance factors out of the sector choice: summing over sectors gives the share who commute to j,

$$\pi_{ij} = \frac{(\mathcal{W}_j/\tau_{ij})^{\theta_L}}{\sum_k (\mathcal{W}_k/\tau_{ik})^{\theta_L}}, \qquad \mathcal{W}_j \equiv \left[\sum_s (w_j^s)^{\theta_L}\right]^{1/\theta_L} \tag{14}$$

where $\mathcal{W}_j$ is the wage index at j, and dividing gives the sector split $\pi_{ij}^s/\pi_{ij} = (w_j^s/\mathcal{W}_j)^{\theta_L}$. The sector composition of the workforce at j is therefore the same for every origin, and the sectoral wage gap at j is pinned down by the sectoral employment split, $w_j^M/w_j^A = (L_j^M/L_j^A)^{1/\theta_L}$. The expected wage is $\bar{w}_i = \sum_j \sum_s \pi_{ij}^s w_j^s$. Workers' expected utility $\bar{v}_i$ is higher, reflecting the value of choosing

their preferred job. Analogous to the price index, it rises with commuting market access $\text{CMA}_i \equiv \left[\sum_j (\mathcal{W}_j/\tau_{ij})^{\theta_L}\right]^{1/\theta_L}$, so that $\bar{v}_i \propto \text{CMA}_i$.

Factor markets and incomes: Firm revenue in sector s in location j is total spending on its goods, $X_j^A = \chi \sum_i \pi_{ij}^{G,A} E_i$ and $X_j^M = (1-\chi)\sum_i \pi_{ij}^{G,M} E_i$, where E_i is income in i and $\pi_{ij}^{G,s}$ its sector-s expenditure share on j. We assume factor markets are perfectly competitive, so revenue is split in fixed Cobb–Douglas shares: agricultural workers earn $w_j^A L_j^A = \alpha X_j^A$ and the landowner earns $r_j H_j = (1-\alpha)X_j^A$, whilst non-agricultural workers take all of X_j^M, so $w_j^M L_j^M = X_j^M$, where $L_j^s = \sum_i L_i \pi_{ij}^s$ is the workforce commuting in to that sector. Land rents come out of agricultural revenue alone, since non-agriculture uses no land.

Welfare: A worker spends labour income on goods and enjoys their housing endowment. Their utility is Cobb–Douglas in consumption and housing services (since housing is a non-tradable endowment, all of the worker's income is spent on consumption goods). The expected utility of a worker resident in i (averaging over their idiosyncratic job preference draws) is therefore

$$V_i^w = \left(\frac{\bar{v}_i}{P_i}\right)^{1-\beta} B_i^{\beta} \tag{15}$$

where $\bar{v}_i$ is the option value of the jobs they can reach, P_i their cost of living, B_i their housing services and β the weight of housing in utility. Worker welfare rises with market access: to well-paying jobs in both sectors (CMA_i) and to cheap food and cheap non-food (GMA_i^A and GMA_i^M). The landowner has no housing and consumes only goods, so their expected utility is $V_i^r = (1-\alpha)X_i^A/P_i$.

6.2 Equilibrium

Given location fundamentals $\{T_i^A, T_i^M, H_i, B_i, L_i\}_{i=1}^N$, parameters $\{\alpha, \beta, \chi, \theta_T, \theta_L, \sigma\}$, and bilateral trade and commuting costs $\{\kappa_{ij}^G, \tau_{ij}\}$, a competitive equilibrium is a vector of factor prices $\{w_j^A, w_j^M, r_j\}_{j=1}^N$ at which agents solve the problems above – pinning down c_j^s, $\pi_{ij}^{G,s}$, P_i^s, P_i, π_{ij}^s, $\bar{w}_i$ and $\bar{v}_i$ – and all markets clear:

(i) Labour: each sector's wage bill equals labour's share of that sector's revenue, $w_j^A L_j^A = \alpha X_j^A$ and $w_j^M L_j^M = X_j^M$, where $L_j^s = \sum_i L_i \pi_{ij}^s$ is the workforce commuting in to that sector.

(ii) Land: rents are pinned down by land's share of agricultural revenue, $r_j = (1-\alpha)X_j^A/H_j$.

(iii) Goods: each sector's revenue equals spending on its output, $X_j^A = \chi \sum_i \pi_{ij}^{G,A} E_i$ and $X_j^M = (1-\chi)\sum_i \pi_{ij}^{G,M} E_i$, where income is $E_i = L_i \bar{w}_i + (1-\alpha)X_i^A$.

We fix the agricultural wage in the first location as the numeraire, $w_1^A = 1$. By Walras's law one market-clearing condition is then redundant.

6.3 A flood in the model

We define a flood as a simultaneous negative shock to agricultural productivity and housing. Locations which are flooded experience a reduction in T_j^A, lowering average productivity in agriculture, and B_j, as houses are destroyed.

In the model, damage to housing has no general-equilibrium consequences. Housing is non-tradable across households or locations – so there is no way that households can act to change their access to housing services in response to damage – and it is separable in utility, so it does not affect their labour supply or consumption decisions. In practice, housing damage may have wider consequences, through reconstruction demand and building-supply prices, but we omit these from the model for now, since the large reconstruction housing programme was only just beginning in 2024, the last time we surveyed our households.

Changes in agricultural productivity, on the other hand, can have impacts that ripple out across the entire economy. They change wages and prices where production takes place, and residents shop and work in places other than where they live. As a result, a resident of i is exposed to prices and wages wherever they buy goods and supply labour. Linearising around the pre-flood equilibrium we can collect these exposures into a single expression.

$$d \ln V_i^w = \underbrace{\beta \, d \ln B_i}_{\text{housing}} + (1-\beta) \Big[\underbrace{\textstyle\sum_j \sum_s \pi_{ij}^s \, d \ln w_j^s}_{\text{wage}} - \underbrace{\textstyle\sum_j \sum_s \chi_s \, \pi_{ij}^{G,s} \, d \ln p_j^s}_{\text{cost of living}} \Big] \tag{16}$$

where $\chi_A = \chi$ and $\chi_M = 1 - \chi$. The change in average welfare for a worker living at i depends on the change in their housing endowment $d \ln B_i$, the change in wages in all locations and sectors $d \ln w_j^s$ weighted by the probability that they held each job prior to the flood π_{ij}^s, and the change in prices in all locations and sectors $d \ln p_j^s$ weighted by the share of their expenditure that was spent on goods from j in sector s prior to the flood.

The landowner holds no housing and lives off rent, so two channels collapse to one:

$$d \ln V_i^r = \underbrace{d \ln r_i}_{\text{rent}} - \underbrace{\textstyle\sum_j \sum_s \chi_s \, \pi_{ij}^{G,s} \, d \ln p_j^s}_{\text{cost of living}}, \qquad d \ln r_i = d \ln X_i^A \tag{17}$$

Together these equations map any flood to its welfare cost at every location.

The wage and price changes are themselves general-equilibrium responses to the flood. To see where they come from, recall that the price in a location is pinned down by unit cost. Let p_j^A be the mill price of an average agricultural good produced in j. Firms are perfectly competitive, so they price at marginal cost – the input bundle c_j^A per unit of output:

$$p_j^A \;\propto\; \frac{c_j^A}{z_j^A}, \qquad c_j^A = (w_j^A)^{\alpha} r_j^{1-\alpha} \tag{18}$$

A flood lowers T_j^A, hence z_j^A: it takes more land and labour to make a unit of output, so unit cost rises. Differentiating, and splitting the bundle cost into its two factors,

$$\underbrace{-\,a_j}_{\text{unit cost up}} \;=\; \underbrace{d\ln p_j^A}_{\text{buyers pay more}} \;-\; \underbrace{\alpha\, d\ln w_j^A}_{\text{workers earn less}} \;-\; \underbrace{(1-\alpha)\, d\ln r_j}_{\text{landowners earn less}} \tag{19}$$

where a_j is the proportional change in average agricultural productivity, negative under the flood. Zero profits imply that a fall in productivity must be absorbed by some mix of a higher price paid by buyers, lower wages and lower land rents.

The trade and commuting elasticities govern how the loss is divided between them. If the trade elasticity is low, the flooded location passes its lower productivity into a higher price, leaving wages and rents largely unchanged. If it is high, the price cannot move, and the split between wages and rents depends on the commuting elasticity: if labour is mobile, the agricultural wage is held up by workers leaving for jobs elsewhere or in non-agriculture, and the residual loads onto land rent. The loss falls on whichever margin cannot adjust.

6.4 Parameter estimation

The trade elasticity and trade costs: We estimate the trade elasticity and the elasticity of trade costs with respect to travel time in two steps. To estimate the product $\theta_T \tau^G$ we run a gravity regression. We parameterise iceberg trade costs as a constant-elasticity function of travel time, $\kappa_{ij}^G = t_{ij}^{\tau^G} e_{ij}$, and defining destination and origin fixed effects λ_i and γ_j and $\eta_{ij} \equiv e_{ij}^{-\theta_T}$, we can write the trade share (12) as:

$$\pi_{ij}^{G,s} = \exp\big(\lambda_i + \gamma_j\big)\, t_{ij}^{-\theta_T \tau^G}\, \eta_{ij} \tag{20}$$

We estimate (20) using internal trade data covering arrivals at all the major markets in Punjab (described in Section 3.6), and travel times by road as described in Section 3. To address zero flows, we estimate by Poisson pseudo-maximum likelihood. Using pooled data for the trade years 2018–19 to 2024–25, we get $\theta_T \tau^G = 1.99$ with a standard error of 0.081.

To separate θ_T from τ^G, we use the fact that in the model the change in log agricultural prices in a location equals $-1/\theta_T$ times the change in its log agricultural market access. We regress the log food price index from our survey on the flood-induced change in agricultural market access implied by the 2022 floods. To address endogeneity, we use an IV approach analogous to that of Section 4. This gives $\theta_T = 2.70$, which implies $\tau^G = 1.99/\theta_T = 0.74$.

The commuting elasticity and commuting costs: We estimate the product $\theta_L \tau^L$ from a gravity regression of commuting flows between hexagons in our survey data on travel time, by Poisson pseudo-maximum likelihood, with a dummy for working in the home hexagon. This gives $\theta_L \tau^L = 2.36$, with a standard error of 0.08. To separate the two, we choose θ_L to match the standard deviation of log wages across workplaces in our survey, 0.59, which gives $\theta_L = 2.79$ and $\tau^L = 0.85$. We calibrate the cost of working outside the home hexagon to match the 77% of workers who work in their own hexagon.

The labour share: We set $\alpha = 0.40$, the share of agricultural revenue paid to workers rather than to landowners. We estimate it from the Agriculture Policy Institute's 2021–22 cost-of-production data, which surveys growers and traders. This is close to the value of 0.46 that Sotelo (2020) uses for Peru.

The housing share: We set $\beta = 0.19$, the share of housing in utility, which governs how much of a flood's welfare cost arises through destroyed housing rather than through prices and wages. We compute it as the housing share of spending among rural households in the 2018–19 Pakistan Social and Living Standards Measurement Survey. This is slightly below typical values in the literature, including the value of 0.25 that Sturm et al. (2023) obtain for Dhaka using the same approach.

The elasticity of substitution across goods: We take $\sigma = 2.4$ from Sotelo (2020). This parameter does not affect our results, since it governs substitution across goods within a sector, and a flood affects every agricultural good in a location equally. What matters instead is substitution across sources, which is governed by θ_T.

6.5 Quantifying the incidence of a flood

Given a map of flooding, the model delivers the change in the average welfare of residents in each location. We compute this in four steps. First, we define the extent of the economy and the unit of a location. Second, we invert the model for the unobserved pre-flood productivities. Third, we estimate the effects of flooding on agricultural productivity and housing. Fourth, we combine a flood map with these elasticities to compute post-flood productivities and housing, and solve for the new equilibrium.

Step 1: Geography. We quantify the model for the province of Sindh, which we divide into hexagons at resolution 7 of the H3 grid, each with an area of approximately 5 km^2 and a radius of about 1.2 km. Each hexagon is a location in the model. The inversion in Step 2 recovers no productivity for 621 hexagons, which either have no agricultural revenue or have an agricultural employment share that no non-negative non-agricultural wage can reproduce given the jobs within commuting reach, and we exclude them, leaving 13,416 locations. We treat Sindh as a closed economy, with no trade or commuting with the rest of Pakistan.

Step 2: Model inversion. We recover the unobserved pre-flood productivities by inverting the model, using data on agricultural revenue X_j^A, residential population L_i and the share of each hexagon's residents working in agriculture (Section 3). Agricultural wages satisfy labour market clearing,

$$w_j^A L_j^A = \alpha X_j^A$$

where the agricultural workforce L_j^A depends on wages everywhere through commuting. Given agricultural wages, the agricultural employment share pins down non-agricultural wages. If few residents work in agriculture despite well-paid agricultural jobs nearby, non-agricultural jobs within reach must pay well. Wages and land rents give each hexagon's income E_i. Productivities then satisfy goods market clearing,

$$X_j^s = \chi_s \sum_i \pi_{ij}^{G,s} E_i$$

so a hexagon that earns high revenue despite high unit costs or poor access to buyers must be productive.

Step 3: Effects of flooding. To simulate a flood in the model, we need to know how it changes each location's fundamentals. By assumption, a flood changes only agricultural productivity and the housing stock, and we estimate the elasticity of each with respect to flood intensity using the 2022 floods. For productivity, we estimate a difference-in-differences regression of agricultural yields, output per km^2 of cropland, on flooding in each hexagon, using data from 2018 to 2022, and use the

change in yields as a proxy for the change in productivity. For housing, we estimate the effect of flooding in each hexagon on the share of its buildings damaged, using the same instrumental-variable approach as in Section 4, and assume that a damaged house loses 95% of its housing services.

Step 4: New equilibrium. For a given flood map, we compute post-flood agricultural productivity and housing in each hexagon from its flood intensity and the estimated elasticities. We then solve for the new equilibrium, holding parameters, trade and commuting costs, and residential population fixed. The incidence of the flood is the resulting change in the average welfare of each hexagon's residents.

7 THE ECONOMIC INCIDENCE OF SINDH'S 2022 FLOODS

We use the model to evaluate the economic incidence of the 2022 floods in Sindh. Panel A of Figure 8 is a map of the 2022 floods in Sindh, measured using the same flood index as in our reduced-form analysis. Two in five hexagons experienced almost no flooding. For the median hexagon, 1.5% of land was under water in September and 0.2% in December, with a maximum depth of 0.15 m. For hexagons at the 75th, 90th and 95th percentiles of the flood index, the corresponding figures were 15%, 43% and 66% in September, 1.7%, 4.8% and 11.8% in December, and 0.58 m, 1.14 m and 1.11 m.

Our estimates imply that these floods reduced average agricultural productivity across Sindh by 14.1% and average housing services by 10.9%. For the median hexagon, productivity fell by 4.7% and housing services by 3.3%. At the 75th, 90th and 95th percentiles, the corresponding falls were 19.1%, 41.9% and 62.2% for productivity and 13.9%, 31.8% and 49.7% for housing services.

Panel B of Figure 8 shows the resulting change in welfare in each hexagon. Welfare fell everywhere, including in hexagons that experienced almost no flooding. The average loss was 12.2%, and the smallest was 1.6%. For the median hexagon, welfare fell by 8.3%, and at the 75th, 90th and 95th percentiles by 14.8%, 24.3% and 34.2%.

Using the model, we can decompose each hexagon's welfare loss into the part due to its own flooding and the part due to flooding elsewhere. Let f_j denote flooding in hexagon j and f_{-j} flooding in all other hexagons, so that welfare in j is $V_j(f_j, f_{-j})$. For each hexagon, we solve the model four times: without the flood, $V_j(0,0)$; applying the 2022 flood map to all hexagons, $V_j(f_j, f_{-j})$; applying it to j only, $V_j(f_j, 0)$; and applying it to all hexagons except j, $V_j(0, f_{-j})$. We can then compute the total effect of the flood on j, $V_j(f_j, f_{-j}) - V_j(0,0)$, the direct effect, $V_j(f_j, 0) - V_j(0,0)$, and the indirect effect, $V_j(0, f_{-j}) - V_j(0,0)$.

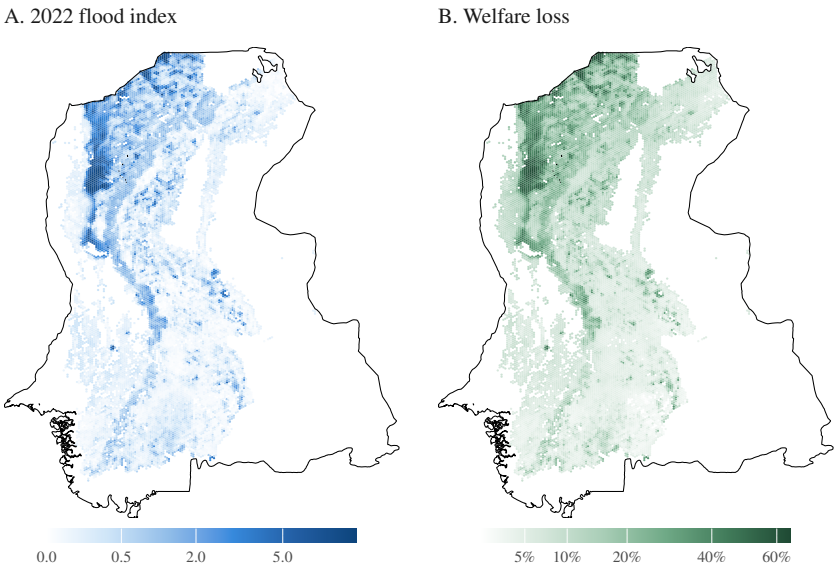


Figure 8: The 2022 floods and the model-predicted welfare loss

Notes: Panel A maps the 2022 flood index in each hexagon, the first principal component of the monthly MODIS and Sentinel-1 inundation shares from July to December and FLEXTH depth, shifted so that the least flooded hexagon takes the value zero. Panel B maps the model-predicted fall in the welfare of each hexagon's residents, including the loss of housing services, as a percent of its no-flood level. The unit of observation is a hexagon of 1.2 km radius. The sample is the 13,416 hexagons covered by the model, drawn from twenty districts of Sindh. Districts outside the model are unfilled.

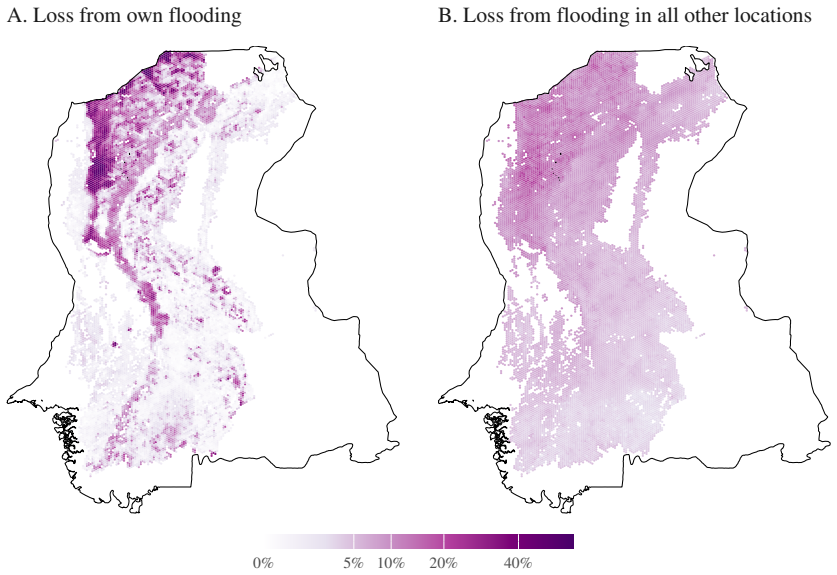


Figure 9: Welfare loss from own flooding and from flooding elsewhere

Notes: Panel A maps the welfare loss each hexagon's residents would have suffered from its own flooding alone, with every other hexagon unflooded, including the loss of housing services. Panel B maps the loss from flooding in every other hexagon, with its own flooding set to zero. Both are percent of the no-flood level. The unit of observation is a hexagon of 1.2 km radius. The sample is the 13,416 hexagons covered by the model, drawn from twenty districts of Sindh. Districts outside the model are unfilled.

Panels A and B of Figure 9 show the direct and indirect effects for each hexagon. The direct effect follows the pattern of flooding and ranges from 0 to 59%. The indirect effect is smoother, higher in the north and lower in the south, and ranges from 0.8 to 25%. For the average hexagon, 73% of the welfare loss is due to flooding elsewhere. This share falls with own flooding, from 99% in the least-flooded decile to 32% in the most-flooded. Since by assumption housing is not indirectly impacted by floods, the share of loss due to indirect exposure is higher for real consumption, at 80% on average, ranging from 99% in the least-flooded decile to 46% in the most-flooded.

8 TARGETING DISASTER ASSISTANCE

We have shown that, for the 2022 floods, most of the welfare loss in the average location was due to flooding elsewhere. Disaster relief, by contrast, is typically targeted on physical exposure, such as flooding or damage to homes.

Do our results imply that relief is misallocated? Not necessarily. First, own flooding may be highly correlated with total loss, in which case targeting on it reaches the right locations anyway. Second, the 2022 floods were unusually large and dispersed, and own flooding may be a better guide to loss in smaller or more concentrated floods. Third, aid may have general-equilibrium effects. Aid that restores production in flooded locations would also reverse the losses those locations impose on others. We consider each in turn.

8.1 Is own flooding a good proxy for need?

Although own flooding accounts for only a minority of the average location's loss, it may still be a good guide to which locations lose most.

8.1.1 Simple targeting rules

A simple first way to assess this is to compare the two rankings. We rank hexagons by their own flooding, measured by the flood index, and by their model-predicted welfare loss. A regression of the loss rank on the flooding rank yields an R^2 of 0.71, so a hexagon's rank in flooding explains 71% of the variation in its rank in welfare loss.

This R^2 is difficult to interpret on its own. A more concrete test compares the allocations implied by simple targeting rules based on physical exposure and on model-predicted welfare loss. After a disaster, governments often declare a list of affected areas and pay a fixed transfer to households

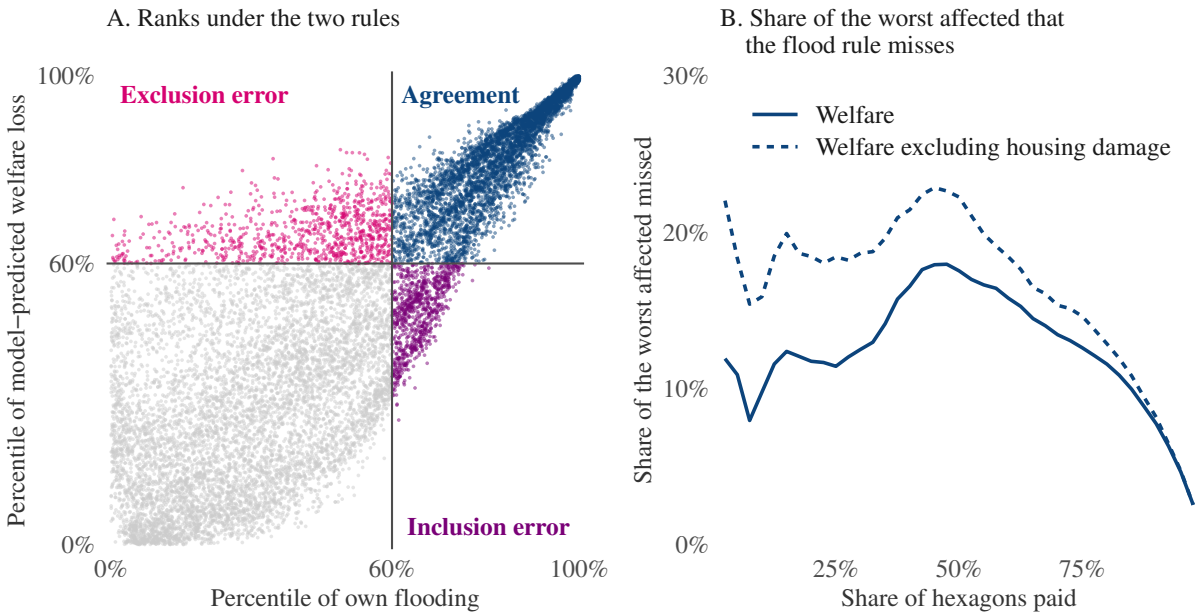


Figure 10: Targeting on own flooding against targeting on model-predicted welfare loss

Notes: Panel A plots each hexagon's percentile of model-predicted welfare loss against its percentile of the 2022 flood index. The lines mark the 60th percentile on each axis, so hexagons above the horizontal line are the 40% with the largest losses and hexagons to the right of the vertical line are the 40% most flooded. Blue points are paid under both rules, pink points are paid only by a rule targeting welfare loss (exclusion errors), and purple points are paid only by a rule targeting flooding (inclusion errors). Panel B plots, for each share of hexagons paid from 2.5% to 97.5%, the share of the hexagons with the largest losses that a rule paying the same share of the most flooded hexagons does not pay. The solid line ranks hexagons by welfare loss and the dashed line by welfare loss excluding housing damage. The unit of observation is a hexagon of 1.2 km radius. The sample is the 13,416 hexagons covered by the model, drawn from twenty districts of Sindh.

living in them. After the 2022 floods, for example, every household below the eligibility threshold of Pakistan's standing cash transfer programme (the Benazir Income Support Programme) that lived in an area notified as flood-affected was entitled to a one-off transfer of 25,000 rupees (Ministry of Finance, Government of Pakistan, 2023; Associated Press of Pakistan, 2022).

Suppose the government could pay a transfer to residents of 40% of hexagons, and was choosing between paying the 40% most flooded and paying the 40% with the largest model-predicted welfare losses. Panel A of Figure 10 plots each hexagon's percentile in welfare loss against its percentile in flooding, with lines marking the top 40%. The two rules agree on 4,479 hexagons (top right). The flood rule would miss the remaining 888 hexagons with the largest model-predicted welfare losses, 17% of the total, because their flooding falls below the cutoff (top left). If the goal were to compensate welfare losses, these would be exclusion errors. The flood rule would instead pay 888

hexagons with flooding above the cutoff but welfare losses below it (bottom right), which would be inclusion errors.

Panel B repeats this comparison for every share of hexagons paid. It plots the share of the hexagons with the largest model-predicted welfare losses that the flood rule would miss. When the government pays 20% of hexagons, the flood rule misses 12% of the worst affected. When it pays half, the flood rule misses 18%. The share falls towards zero beyond this point, because both rules then pay almost every hexagon. Excluding housing losses, which are driven by own flooding by construction, the flood rule misses 18% and 22% of the worst affected. Own flooding is therefore a better guide to loss among the most flooded hexagons than among the moderately flooded, and even a narrowly targeted flood rule misses around one in eight of the worst affected.

Figure 11 maps the two types of error. The exclusion errors are less-flooded hexagons in heavily flooded surroundings. The median hexagon in this group lies at the 47th percentile of own flooding but the 68th percentile of flooding in other hexagons within 10 km. Flooding elsewhere accounts for 92% of its welfare loss. The inclusion errors are the opposite case, at the 65th and 44th percentiles, and flooding elsewhere accounts for 58% of their welfare loss.

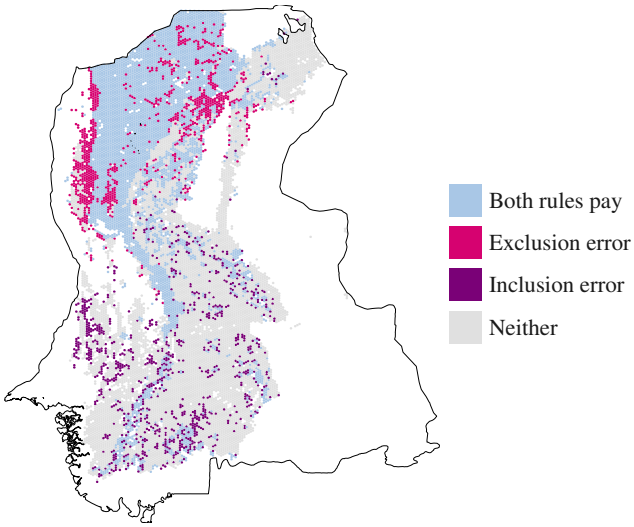


Figure 11: Inclusion and exclusion errors under the flood rule

Notes: Consider two rules for allocating aid, one paying the 40% of hexagons with the highest 2022 flood index and one paying the 40% with the largest model-predicted welfare loss. Blue hexagons are paid under both rules. Pink hexagons are paid only under the welfare-loss rule, and so are exclusion errors of the flood rule. Purple hexagons are paid only under the flood rule, and so are inclusion errors. Grey hexagons are paid under neither. The unit of observation is a hexagon of 1.2 km radius. The sample is the 13,416 hexagons covered by the model, drawn from twenty districts of Sindh. Districts outside the model are unfilled.

8.1.2 Optimal targeting

To measure the welfare cost of inclusion and exclusion errors, we must make an assumption about social welfare. We set out the planner's problem, and use it to quantify the value of providing the planner with our model-based estimates of welfare loss.

We assume that the planner cares about the welfare losses caused by the flood, and places more weight on larger losses. The planner can make transfers to households in different locations from a fixed budget, and allocates them to minimise the sum, across households, of an increasing and convex function of each household's welfare loss. Let $V_{ih}(\mathbf{f}, \mathbf{y})$ denote the welfare of household h in hexagon i, given flooding $\mathbf{f}$ and transfers $\mathbf{y}$ in all hexagons. The planner solves

$$\min_{\{y_{ih} \geq 0\}} \sum_i \sum_h \omega_{ih}\, g\big(V_{ih}(\mathbf{0}, \mathbf{0}) - V_{ih}(\mathbf{f}, \mathbf{y})\big) \quad \text{s.t.} \quad \sum_i \sum_h y_{ih} = B \tag{21}$$

where $\omega_{ih} > 0$ are welfare weights, g is strictly increasing and strictly convex, and the budget B is financed from outside the economy.

The planner cares about losses rather than levels of welfare, reflecting the apparent aim of disaster relief to compensate households for what they lost rather than to reduce poverty. For example, post-disaster needs assessments organise assistance around damage and loss (World Bank et al., 2013), and US federal disaster assistance has no income test (Webster, 2023). The two objectives imply the same allocation when pre-flood welfare is equal across households. Otherwise, the loss-based planner places more weight on richer households, who have more to lose.

If the planner observes each household's welfare loss, the optimal transfers satisfy

$$\sum_k \sum_{h'} \omega_{kh'}\, g'\big(V_{kh'}(\mathbf{0}, \mathbf{0}) - V_{kh'}(\mathbf{f}, \mathbf{y})\big)\, \frac{\partial V_{kh'}}{\partial y_{ih}} = \mu - \lambda_{ih}, \qquad \lambda_{ih} \geq 0, \quad \lambda_{ih}\, y_{ih} = 0 \tag{22}$$

where μ is the multiplier on the budget constraint and λ_{ih} is the multiplier on the non-negativity constraint. The social value of a transfer to household h in hexagon i is the reduction in losses it causes in every hexagon, valued at g'.

If transfers are made in units of utility, so that a transfer of y_{ih} raises the welfare of its recipient by y_{ih}, and transfers have no general-equilibrium effects, then equation 22 simplifies to

$$\omega_{ih}\, g'\big(V_{ih}(\mathbf{0}, \mathbf{0}) - V_{ih}(\mathbf{f}, \mathbf{y})\big) = \mu - \lambda_{ih} \tag{23}$$

With equal welfare weights, the planner gives nothing to households whose losses fall below a cutoff and reduces the losses of all other households to that cutoff, with the cutoff set to exhaust the budget.

We use this setup to compare three allocations of the same budget. In each, the planner sets a per-household transfer for each hexagon. In the first, this transfer is proportional to the hexagon's own flooding. In the second, it depends only on own flooding, but the planner chooses the schedule optimally using the predicted welfare losses. In the third, the planner sets it using the hexagon's predicted welfare loss, according to equation 23. Comparing the first and second allocations isolates the value of information about welfare losses. Comparing the second and third isolates the value of conditioning transfers on that information.

Take, for example, a planner with $g(\ell) = \ell^{1+\varepsilon}/(1+\varepsilon)$ and $\varepsilon = 0.5$, and a budget equal to 25% of the total welfare loss from the flood. Transfers proportional to own flooding reduce the social welfare loss by 37%. This exceeds 25% because g is convex and these transfers go mostly to households with large losses. Choosing the schedule optimally, but still conditioning only on own flooding, achieves the same reduction with 97% of the budget. Conditioning transfers on predicted welfare losses achieves it with 82%. Almost all of the gain therefore comes from conditioning transfers on predicted losses, not from using them to design the schedule.

Figure 12 shows how these savings vary with the budget and with ε. Savings are smallest at a budget of around 25% of the total loss, and larger at both smaller and larger budgets (Panel A). At small budgets, both optimal allocations gain by concentrating transfers on the largest losses. At large budgets, targeting on predicted losses also avoids overpaying the most flooded hexagons and reaches those whose losses are due to flooding elsewhere. The gap between the two optimal allocations, the value of conditioning transfers on the predicted losses, is 12 to 15 percentage points at every budget. This value is zero when g is linear and rises with the planner's aversion to large losses, to 25 percentage points at $\varepsilon = 1$ and 42 at $\varepsilon = 3$ (Panel B).

8.2 Other floods

Our analysis so far has focused on a single case study, a large and spatially dispersed flood in a low-income economy. To extrapolate from our findings to other disasters and economies, and to draw broader lessons for the design of disaster assistance, we use the model to study how our results depend on features of the flood and of the economy. We focus on three: how integrated the economy was before the flood, the spatial concentration of the flood and its size.

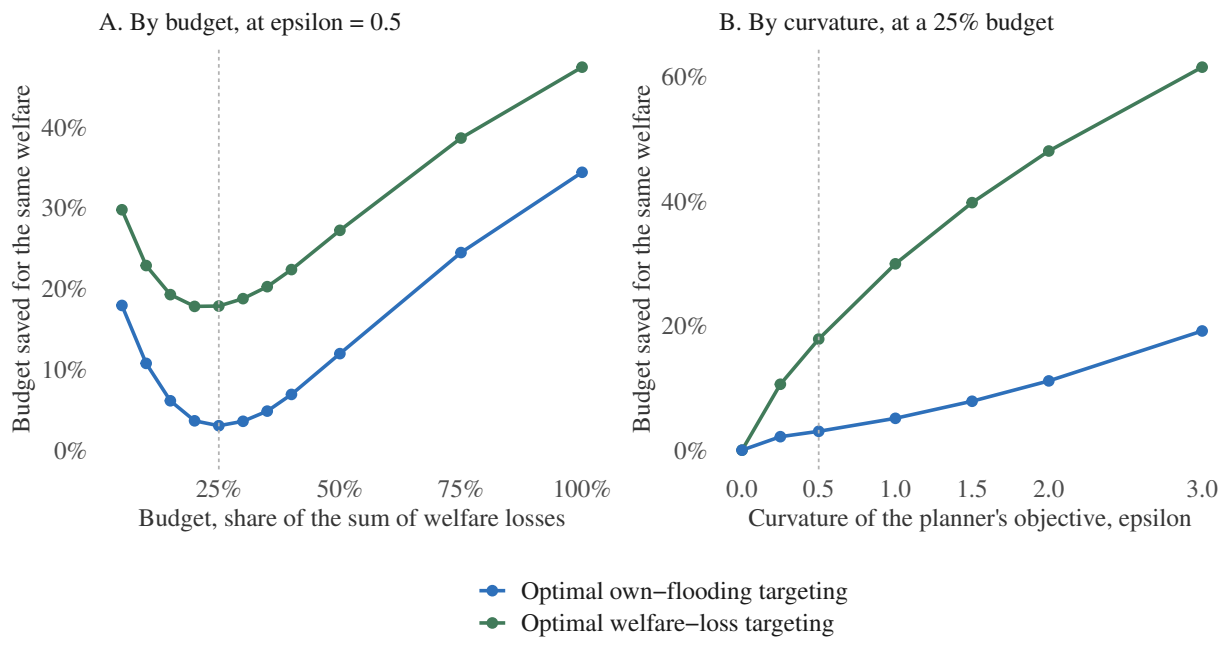


Figure 12: Budget saved by optimal targeting, by budget and by curvature of the planner's objective

Notes: Take the social welfare that transfers proportional to own flooding achieve with a given budget. Each point is the share of that budget an alternative targeting rule could save whilst achieving the same social welfare. The planner minimises the sum across residents of $g(\ell) = \ell^{1+\varepsilon}/(1+\varepsilon)$, where ℓ is the resident's welfare loss after transfers, and every resident of a hexagon receives the same transfer in units of utility. The blue line conditions transfers on own flooding alone, with the schedule chosen optimally. The green line conditions them on model-predicted welfare loss. Panel A varies the budget, expressed as a share of the sum of residents' welfare losses, with $\varepsilon = 0.5$. Panel B varies ε with the budget at 25% of that sum. Dashed lines mark the values used in the text. The unit of observation is a hexagon of 1.2 km radius. The sample is the 13,416 hexagons covered by the model, drawn from twenty districts of Sindh.

8.2.1 Pre-flood economic integration

We have argued that the incidence of a flood differs from its physical footprint because locations are linked through trade and commuting, and these economic linkages share the costs of a disaster across space. It follows that the costs of trade and commuting are an important determinant of a flood's incidence.

To see why, consider an economy in which these costs are high, so that residents mostly work and shop where they live (π_{ii} and π^G_{ii} close to one). These shares also barely respond to wages and prices, since the elasticities of π_{ii} with respect to the local wage and of π^G_{ii} with respect to the local price are $\theta_L(1-\pi_{ii})$ and $-\theta_T(1-\pi^G_{ii})$. Three implications follow. First, by equation 16, residents of i are exposed to first order only to changes in wages and prices in i. Second, flooding in other locations affects wages and prices in i only through changes in labour supply to i or in demand for its goods. Since commuting and trade shares are unresponsive, both are approximately unchanged, and

wages and prices in i depend only on flooding in i. Third, residents of i cannot substitute away from these changes by working or buying elsewhere. The welfare loss from a flood is therefore borne by the locations it floods.

By contrast, when trade and commuting costs are low, residents work and shop widely, so π_{ii} and π_{ii}^{G} are small and highly responsive to wages and prices. Residents of i are then exposed to wages and prices in many locations, each of which depends on flooding everywhere, because workers and buyers reallocate towards wherever wages are higher and prices lower. The productivity loss from a flood is therefore shared across the whole economy.

8.2.2 Spatial dispersion of the flood

Holding the share of locations flooded fixed, the spatial pattern of a flood changes who bears its cost. Since π_{ij} and π_{ij}^{G} fall with distance, residents of i work and shop mainly in locations close to i, and the spillovers from flooding in i fall mainly on those locations. In a concentrated flood, flooded locations tend to have flooded neighbours and unflooded locations unflooded neighbours. To first order, flooded locations are therefore exposed to flooding both directly and indirectly, while unflooded locations are exposed to neither. Moreover, residents of flooded locations cannot offset these changes by shifting their labour supply or goods demand elsewhere, because the locations they can reach are flooded too.

In a dispersed flood, the locations close to a flooded location are dry. Locations tend to be exposed either directly or indirectly, but not both. Moreover, residents of flooded locations can reach unflooded locations in which to supply labour and buy goods, and so share their loss with them. It follows that own flooding is a good predictor of welfare loss in a concentrated flood, when direct and indirect exposure are strongly correlated, and a worse one in a dispersed flood.

We consider counterfactual floods that keep the same distribution of flooding across hexagons as in 2022, but rearrange it so that flooding is more or less spatially concentrated. In the most dispersed flood, unflooded hexagons bear 30% of the total welfare loss, compared with 24% in the most concentrated. The rank R^2 between own flooding and welfare loss rises only slightly, from 0.67 to 0.71.

8.2.3 Size of the flood

In this model, the aggregate cost of a flood is increasing in the number of locations flooded, holding the intensity of flooding fixed. To first order, by equation 16, the aggregate loss is linear in the

number of locations flooded, since trade and commuting redistribute the loss across locations without changing its total. Taking into account second-order effects, residents of flooded locations offset part of their loss by shifting labour supply and goods demand towards unflooded locations, and their ability to do so is greatest when few locations are flooded. The aggregate cost is therefore mildly convex, and more so the lower are trade and commuting costs and the higher are θ_L and θ_T. In autarky, there is no substitution, and it is linear.

The size of a flood also changes its incidence. Linearising around the pre-flood equilibrium, we can write the change in welfare in the form

$$d \ln V_i^w = \beta \, d \ln B_i + \sum_k F_{ik} \, a_k \tag{24}$$

where F_{ik} is the general-equilibrium effect of a productivity shock in k on welfare in i, which depends on the pre-flood trade and commuting shares and on the model's parameters, and $\sum_k F_{ik} = 1 - \beta$.

Consider a flood that lowers productivity by a and housing by b in a set of locations $\mathcal{F}$. The welfare loss in i, to first order, is

$$-d \ln V_i^w = \beta b \, \mathbf{1}\{i \in \mathcal{F}\} + a \sum_{k \in \mathcal{F}} F_{ik} \tag{25}$$

The first term is the loss due to housing destruction, which is the same (βb) for every flooded location regardless of the scale of the flood. The second is the loss due to productivity impacts, which is increasing in the number of flooded locations, whether or not i is itself flooded. As the flood grows, the second term rises in every location while the first does not. The ratio of the loss of a resident of a flooded location to that of a resident of an unflooded location therefore falls. It falls more slowly in a concentrated flood, since newly flooded locations then border locations that are already flooded, and so the productivity impacts fall mainly on locations that are already flooded.

The implication is that own flooding predicts welfare loss less well as the flood grows, up to a point. Own flooding determines the first term in equation 25, so what matters is how much the second term varies across locations. It is zero in every location when none is flooded, and equal to $a(1 - \beta)$ in every location when all are flooded, so it varies most at intermediate sizes. Own flooding therefore ranks welfare losses well in small floods and in very large floods, and worst in between.

Consider counterfactual floods that vary the share of hexagons flooded, holding the distribution of flooding among flooded hexagons and spatial concentration fixed at their 2022 values. The welfare loss per resident of a flooded hexagon is 7.1%, 8.4%, 9.9% and 11.3% when 5%, 20%, 40% and 60%

of hexagons are flooded, and per resident of an unflooded hexagon 1.9%, 3.1%, 4.5% and 5.5%, so the ratio falls from 3.8 to 2.1. The rank R^2 between own flooding and welfare loss is 0.79 for the smallest floods, falls to 0.66 when 30% of hexagons are flooded, and rises slightly thereafter.

8.2.4 Summary

Own flooding is therefore a worse guide to welfare loss when floods are large and dispersed and when trade and commuting costs are neither very high nor very low. Many disasters are much smaller than the 2022 floods, and our results suggest that when a disaster affects few locations, physical exposure is a good guide to welfare loss. However, large floods are projected to become more frequent with climate change, meaning that accounting for indirect exposure will become increasingly important. Our results on trade and commuting costs also suggest that the spillovers we document are not specific to a low-income setting. Trade and commuting costs are typically lower in high-income economies, and our model would predict that a larger share of a flood's cost is then borne outside the flooded locations.

8.3 General-equilibrium effects of aid

The planner's problem above treats aid as a lump-sum transfer of utility, which does not change household behaviour and so has no general-equilibrium effects. Aid that restores productive capacity in the locations where it is spent does have such effects, and the optimal allocation then depends on the spillovers from that spending.

We have so far considered lump-sum transfers of utility. However, it is possible that disaster assistance could restore productive capacity, by draining fields, repairing irrigation or replacing livestock and machinery. Aid of this kind would reverse, at least partially, the productivity loss in the location where it is spent, and so reverse the losses that location imposed on others.

We model this kind of aid as spending y_j in location j that raises agricultural productivity T_j^A. The planner solves equation 21 with y_j in place of utility transfers. We assume for now that aid cannot increase productivity in a location above its pre-flood level ($T_j^A(f_j, y_j) \leq \bar{T}_j^A$).

The first-order condition, equation 22, becomes

$$\frac{\partial \ln T_j^A}{\partial y_j} \sum_i \omega_i \, g'(\ell_i) \, \frac{\partial V_i}{\partial \ln T_j^A} = \mu - \lambda_j + \nu_j, \qquad \nu_j \geq 0, \quad \nu_j \left(\bar{T}_j^A - T_j^A\right) = 0 \tag{26}$$

where $\ell_i = V_i(\mathbf{0}, \mathbf{0}) - V_i(\mathbf{f}, \mathbf{y})$ is the remaining loss in i and ν_j is the multiplier on the cap.

The left-hand side of equation 26 is the social value of a rupee spent in j. An extra rupee in j raises productivity there by $\partial \ln T_j^A / \partial y_j$. This raises welfare in location i by $\partial V_i / \partial \ln T_j^A$ per unit of productivity. The planner values an increase in V_i more highly if i has lost more given the flood and the aid so far, as measured by $\omega_i g'(\ell_i)$.

The planner spends more in j if more households are exposed to its productivity (for example, j is a large employer or food supplier), or if those households have lost more as a result of the flood (for example, they suffered direct damage or sit in an area of widespread flooding). Intuitively, the value of aid in j is its contribution to the market access of all locations, weighted by the losses suffered by residents of these locations. Aid therefore does not necessarily flow to the most damaged locations, since restoring productivity in the locations they trade with and commute to can reduce their losses by more.

Given the constraint that aid can only restore productivity to pre-flood levels, it is allocated only to flooded locations. If this constraint is relaxed, the planner may direct some aid to locations that were not flooded but are highly connected to flooded locations.

9 CONCLUSION

This paper shows that the economic incidence of a natural disaster can differ substantially from its physical footprint. Using original surveys of 5,100 rural households after Pakistan's 2022 floods and a recentered instrument based on a hydrological model, we find that flooding in the surrounding area lowers household consumption, holding flooding at home fixed. Flooding more than 5 km from the home accounts for nearly two-thirds of the effect of flooding within 15 km. The same flooding reduces wage employment and raises food prices, consistent with spillovers through labour and product markets.

To quantify incidence in general equilibrium, we build a spatial model of trade and commuting. For the 2022 floods in Sindh, the model implies that welfare fell in every location, although two in five experienced almost no flooding, and that 73% of the average loss was due to flooding elsewhere. Targeting aid on predicted losses rather than own flooding would have achieved the same welfare at 18% lower cost. The gap between physical exposure and incidence is largest for large, dispersed floods.

The broader lesson is that trade and commuting with unflooded areas insure households against localised shocks, and that this insurance breaks down when the shock is common to the markets and neighbours a household would otherwise turn to. As climate change makes large floods more

frequent, and a single event covers more of a country at once, ignoring the divergence between incidence and physical footprint will increasingly lead us to underestimate the impact of disasters and to misallocate aid.

APPENDICES

A INTERPRETATION OF RELATIVE EFFECTS

The results in Section 5 provide descriptive evidence that spillovers matter. We imposed as little structure as possible to obtain them, remaining agnostic about the channels through which spillovers operate. It would be useful, however, to give these estimates a more structural interpretation, and to know whether they can guide the targeting of aid. For that we need to understand how the predicted values from our regressions relate to the total loss in each location caused by the entire flood.

In this appendix we set out a potential outcomes framework that relates these predicted values to the total loss, and use it to make four points. First, comparisons of flooded and unflooded locations will likely underestimate the total effects of floods and overstate the direct impacts. Second, even our estimates, which partially account for spillovers, might be contaminated by omitted spillovers from flooding beyond 15 km, although this contamination partly helps us predict the loss in each location. Third, limited power forces us to use coarse distance bands, so our estimates capture weighted-average effects within each band. Fourth, our estimates can guide the targeting of aid within districts but not across them, and estimating the total loss in each location requires a model.

We adapt our framework from Hudgens and Halloran (2008).[7] There are many locations, indexed i, each of which is either flooded ($f_i = 1$) or not ($f_i = 0$), though we can easily generalise to the case of continuous flood intensity. Write the vector of flooding in all locations as $\mathbf{f} = (f_i, f_{-i})$, where f_{-i} is flooding in all locations except i. Let $Y_i(\mathbf{f})$ denote an outcome, say consumption, of a household living in i as a function of the whole vector of flooding, abstracting from household-level heterogeneity within a location. Define the direct effect of flooding on i as

$$\mathrm{DE}_i(f_{-i}) \equiv Y_i(1, f_{-i}) - Y_i(0, f_{-i}), \tag{27}$$

the change in its consumption caused by its own flooding, holding fixed flooding everywhere else. Define the spillover or indirect effect on i as

$$\mathrm{IE}_i(f_{-i}) \equiv Y_i(0, f_{-i}) - Y_i(0, \mathbf{0}), \tag{28}$$

[7] Chodorow-Reich (2020) uses a similar framework to relate estimates from regional comparisons to aggregate effects.

the impact on consumption in i of flooding in all locations except i, when i is not flooded. The total effect of the flood $\mathbf{f}$ on a flooded location is the sum of the direct and indirect effect,

$$\mathrm{TE}_i(f_{-i}) \equiv Y_i(1, f_{-i}) - Y_i(0, \mathbf{0}) = \mathrm{DE}_i(f_{-i}) + \mathrm{IE}_i(f_{-i}). \tag{29}$$

Many studies of disasters compare affected with unaffected locations. In this framework, the difference in expected consumption between flooded and unflooded locations is

$$\begin{aligned}\mathbb{E}[Y_i \mid f_i = 1] - \mathbb{E}[Y_j \mid f_j = 0] = \underbrace{\mathbb{E}[\mathrm{DE}_i \mid f_i = 1]}_{\text{direct effect}} + \underbrace{\mathbb{E}[\mathrm{IE}_i \mid f_i = 1] - \mathbb{E}[\mathrm{IE}_j \mid f_j = 0]}_{\text{difference in spillovers}} \\ + \underbrace{\mathbb{E}[Y_i(0, \mathbf{0}) \mid f_i = 1] - \mathbb{E}[Y_j(0, \mathbf{0}) \mid f_j = 0]}_{\text{selection}}\end{aligned} \tag{30}$$

the expected direct effect on flooded locations, plus the difference in expected spillovers between flooded and unflooded locations, plus selection bias. Selection bias is zero if flooding is randomly assigned, an assumption we maintain for the purposes of this section.

If the spillover is the same at every location, as in Nakamura and Steinsson (2014), Chodorow-Reich (2019, 2020) and Wolf (2023), the comparison recovers the direct effect. Adão et al. (2026) show that this holds only when locations respond symmetrically to shocks and there are no spatial frictions. With spatial frictions, spillovers are larger from nearby locations. Suppose that own and nearby flooding both lower consumption. Since flooding is spatially correlated, flooded locations have more flooded neighbours, and so suffer larger spillovers than unflooded locations. The relative estimate therefore lies between the direct effect and the total effect on flooded locations, since it includes only the part of the spillover onto flooded locations that exceeds the spillover onto unflooded locations.

Our regressions do not compare flooded with unflooded locations. Instead, they compare locations within the same district that differ in mean flooding within c km and from c to 15 km. Since these regressors average flooding over many locations, it helps to take a linear approximation of $Y_i(\mathbf{f})$ to interpret their coefficients. To first order,

$$Y_i(\mathbf{f}) = Y_i(\mathbf{0}) + \sum_j b_{ij} f_j, \tag{31}$$

where f_j is now the continuous flood index, $Flood_j$ in Section 4, and b_{ij} is the marginal effect of flooding in j on consumption in i. Under this approximation the direct effect is $\mathrm{DE}_i = b_{ii} f_i$, the spillover is $\mathrm{IE}_i = \sum_{j \neq i} b_{ij} f_j$, and the total loss is $\mathrm{TE}_i = \sum_j b_{ij} f_j$.[8]

Splitting this into terms by distance from i, we can compare it to our regression specification from Section 4 (abstracting from the IV for now),

$$Y_i = \sum_{j:\, d_{ij} \leq c} b_{ij} f_j \quad + \sum_{j:\, c < d_{ij} \leq 15} b_{ij} f_j + \sum_{j:\, d_{ij} > 15} b_{ij} f_j + Y_i(\mathbf{0}) \tag{32}$$

$$Y_i = \beta_{\text{own}} F_i^{\text{own}}(c) + \beta_{\text{near}} F_i^{\text{near}}(c) \quad + \alpha_{D(i)} + \varepsilon_i \tag{33}$$

where d_{ij} is the straight-line distance in km between i and j. To understand what the coefficients we estimate from our regression, $\hat{\beta}_{\text{own}}$ and $\hat{\beta}_{\text{near}}$, capture, and how the predicted loss in each location, $\hat{\beta}_{\text{own}} F_i^{\text{own}}(c) + \hat{\beta}_{\text{near}} F_i^{\text{near}}(c)$, relates to the total loss, TE_i, consider a set of strong assumptions, which we will subsequently relax.

Benchmark. Suppose that flooding more than 15 km away does not affect i ($b_{ij} = 0$ if $d_{ij} > 15$), and that flooding in every location within a distance band has the same effect on i ($b_{ij} = b^{\text{own}}$ if $d_{ij} \leq c$ and $b_{ij} = b^{\text{near}}$ if $c < d_{ij} \leq 15$). Equation 32 then becomes

$$Y_i = N^{\text{own}} b^{\text{own}} F_i^{\text{own}}(c) + N^{\text{near}} b^{\text{near}} F_i^{\text{near}}(c) + Y_i(\mathbf{0}) \tag{34}$$

where N^{own} and N^{near} are the numbers of locations in each band. Without the district fixed effects, $\hat{\beta}_{\text{own}}$ recovers the effect of flooding a single location within c km of i, multiplied by the number of locations within that radius, $N^{\text{own}} b^{\text{own}}$, and likewise $\hat{\beta}_{\text{near}}$ recovers $N^{\text{near}} b^{\text{near}}$. The predicted loss for i, $\hat{\beta}_{\text{own}} F_i^{\text{own}}(c) + \hat{\beta}_{\text{near}} F_i^{\text{near}}(c)$, then equals the total effect of the flood on i, TE_i. These assumptions are, however, highly restrictive. We relax them in turn, first allowing flooding beyond 15 km to affect i, and then allowing locations within a band to differ in their effect on i.

Omitted spillovers: constant within districts. If flooding more than 15 km away does affect i, but this effect is constant across locations within a district, then it will be absorbed by the district fixed effects, leaving the estimated coefficients unchanged. But the predicted loss, $\hat{\beta}_{\text{own}} F_i^{\text{own}}(c) + \hat{\beta}_{\text{near}} F_i^{\text{near}}(c)$, now leaves out this additional spillover, and so understates the total effect of the flood

[8] By linearising around the pre-flood equilibrium, we can write the model of Section 6 in this form, where b_{ij} depends on pre-flood trade and commuting shares and on the elasticities of trade and labour supply (Adão et al., 2026; Kleinman et al., 2024).

on each location i if the spillover lowers consumption.[9] Since this missing component does not vary within a district, the resulting predictions will correctly rank locations by total loss within a district, but without knowing this district-level spillover we cannot compare across districts. In other words, we could use this information for targeting funds within a district, but not for determining the total amount of funds to flow to each district.

OMITTED SPILLOVERS: VARYING WITHIN DISTRICTS. If flooding more than 15 km away does affect i, and this effect varies across locations within a district, then only its district average will be absorbed by the district fixed effects, and the remainder will enter the error term. If this omitted flooding is correlated with flooding within 15 km, as is likely since flooding is spatially correlated, part of its effect will be picked up by $\hat{\beta}_{\text{own}}$ and $\hat{\beta}_{\text{near}}$, which would then over- or understate the effect of flooding within 15 km, depending on the direction of the spillover. From the perspective of ranking locations by their loss, this bias is not a problem, and may even be useful, since it means that our estimates capture the part of the impact of flooding beyond 15 km that can be predicted by flooding within 15 km, which is part of the total loss we want to measure. The predicted loss for each location, $\hat{\beta}_{\text{own}} F_i^{\text{own}}(c) + \hat{\beta}_{\text{near}} F_i^{\text{near}}(c)$, still misses spillover effects that cannot be predicted by flooding within 15 km, or that are common to all locations in a district. As when spillovers are constant within districts, we therefore cannot reliably use the estimates to rank locations across districts. In addition, if the unpredictable part is large, there will be heterogeneity in losses within a district that is not captured by our predictions.

FUNCTIONAL FORM. Our regression imposes a specific functional form, in which flooding in every location within a distance band has the same effect on i. In practice, locations within a band may differ in their effect on i, for example because closer locations matter more. Each coefficient then estimates a weighted average of the effects of the locations in its band, with weights determined by the typical spatial pattern of flooding. The regression gives the best linear predictor of the total loss that uses only mean flooding in each band, given the floods we observe. It would predict less well for a flood with a different spatial pattern, which limits how far our estimates generalise to other floods.

Taken together, these results suggest that our reduced-form estimates are reasonably informative from a targeting perspective. Even when omitted spillovers bias our coefficients, the bias tends to work in favour of our overall objective of predicting the loss in each location, since the coefficients

[9] We cannot simply add the district fixed effect back to the predicted loss, since it contains both this spillover and the district mean of consumption without a flood, $Y_i(\mathbf{0})$.

pick up the part of the omitted spillovers that flooding within 15 km can predict. But the estimates cannot reliably rank locations across districts, and our power limits how flexibly we can let the loss vary with flooding, so that we have to rely on coarse distance bands. None of these problems arises from non-random flooding. Throughout this section we have assumed that flooding is randomly assigned, which is what our IV strategy is designed to address, so these problems affect our IV estimates too.

To estimate the total loss in each location, we therefore need a model. Taking a stance on the channels through which spillovers operate allows us to predict the loss more flexibly, as a function of flooding in all locations rather than only in the distance bands we have the power to estimate, and to capture spillovers from beyond 15 km, including how they vary across space and across districts.

B ADDITIONAL FIGURES

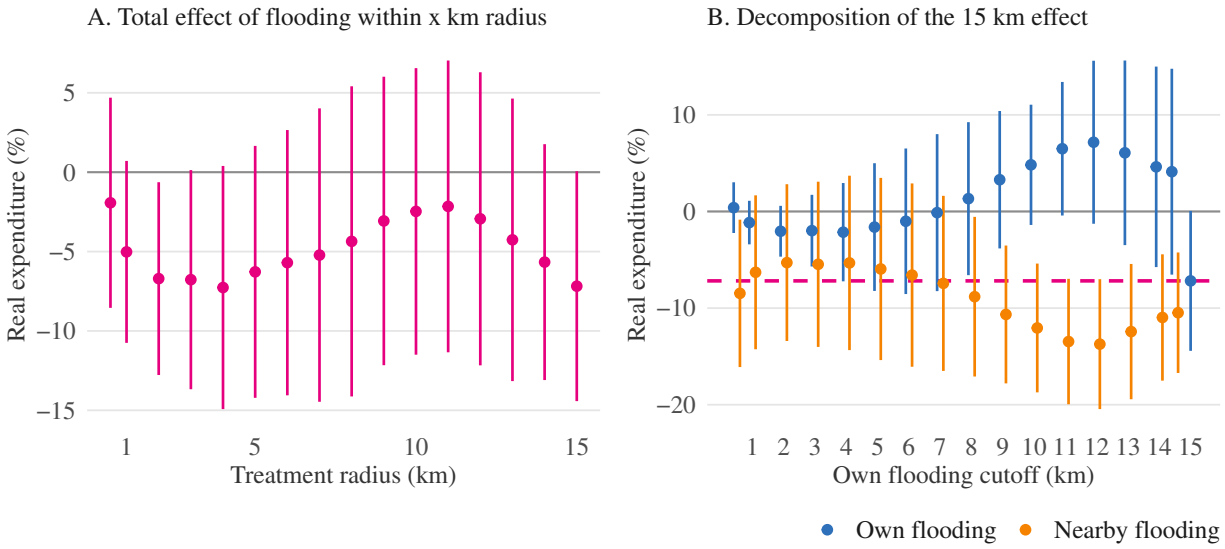


Figure 13: Effect of flooding on real expenditure, households reporting relief

Notes: All estimates are the effect of flooding for households reporting flood relief, the sum of the flooding coefficient and its interaction with a relief indicator, from one regression per point on all households that includes flooding, the indicator and their interaction, instrumented by the recentered hydrological prediction and its interaction. Panel A plots this coefficient for flooding within each radius, multiplied by the within-district difference between the 10th and 90th percentiles of that regressor. Panel B plots, for each cutoff, the coefficients on flooding within the cutoff (blue) and flooding from the cutoff to 15 km (orange) from one regression with both regressors. Each coefficient in Panel B is multiplied by the average change in its regressor when flooding within 15 km moves from its 10th to its 90th percentile within districts. The dashed line marks the 15 km estimate from Panel A, which is also the point at 15 km. The outcome is log monthly household expenditure per adult equivalent, deflated by a household food price index, so estimates are in percent. The unit of observation is a household surveyed in both waves for whom we observe a food price index and whether they received flood relief (N = 4,178). Whiskers are 90% confidence intervals from Conley standard errors with a Bartlett kernel and a cutoff set from the instrument's spatial correlogram. All regressions include district fixed effects.

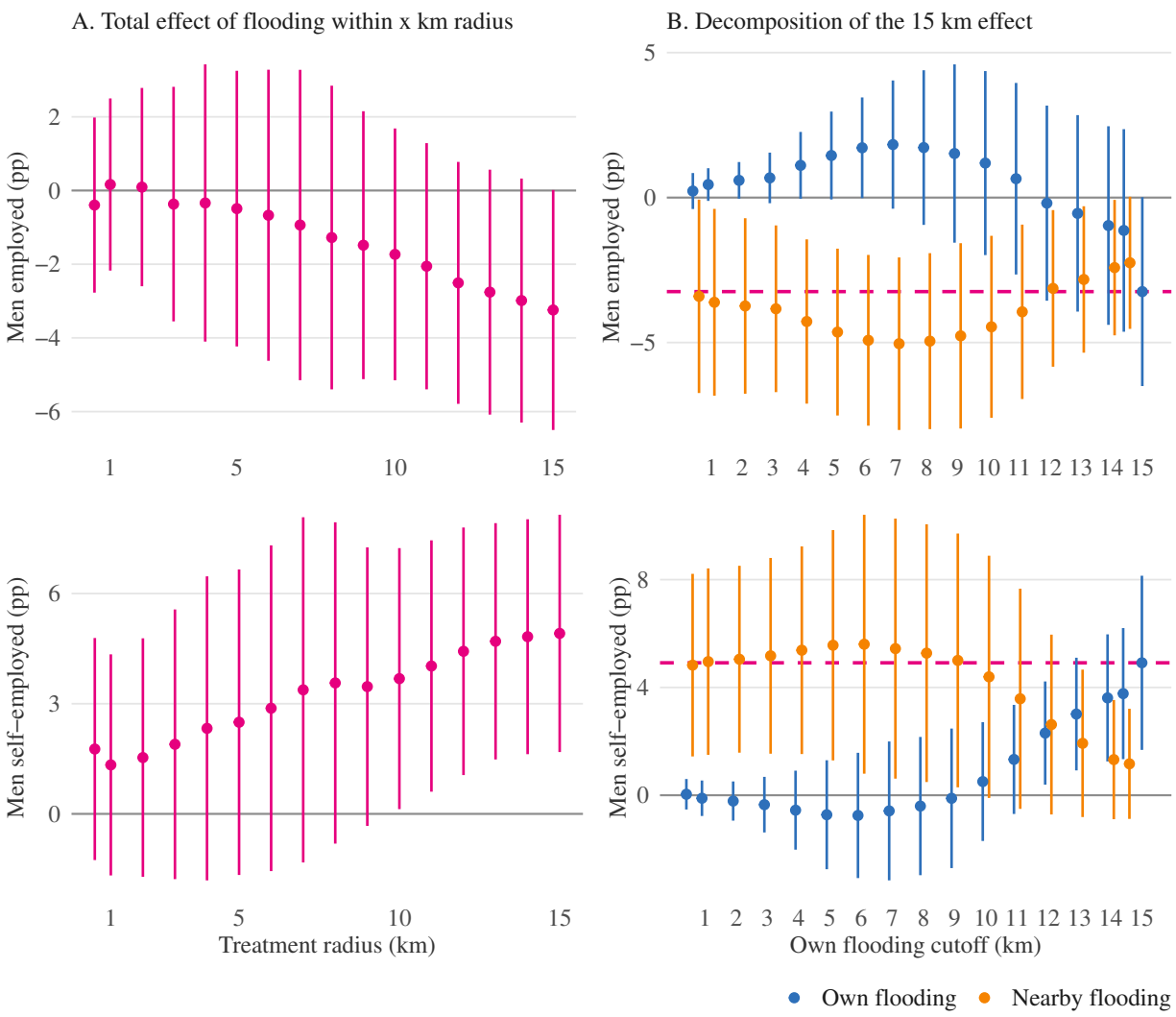


Figure 14: Effect of flooding on employment and self-employment

Notes: Panel A plots the two-stage least squares coefficient on flooding within each radius, instrumented by the mean recentered hydrological prediction over the same radius, multiplied by the within-district difference between the 10th and 90th percentiles of that regressor. Panel B plots, for each cutoff, the coefficients on flooding within the cutoff (blue) and flooding from the cutoff to 15 km (orange) from one regression with both regressors, each instrumented by the recentered prediction over the same area. Each coefficient in Panel B is multiplied by the average change in its regressor when flooding within 15 km moves from its 10th to its 90th percentile within districts. The dashed line marks the 15 km estimate from Panel A, which is also the point at 15 km. The outcome in the top row is the share of the household's men aged 18 to 65 in any employment, controlling for the number of such men. The outcome in the bottom row is the share of the household's men aged 18 to 65 who are self-employed, controlling for the number of such men. In all panels the unit of observation is a household surveyed in both waves, and the sample is households with at least one man aged 18 to 65 whose pre-flood sector we observe (N = 4,478). Whiskers are 90% confidence intervals from Conley standard errors with a Bartlett kernel and a cutoff set from the instrument's spatial correlogram. All regressions include district fixed effects.

B.1 Wage employment by pre-flood sector

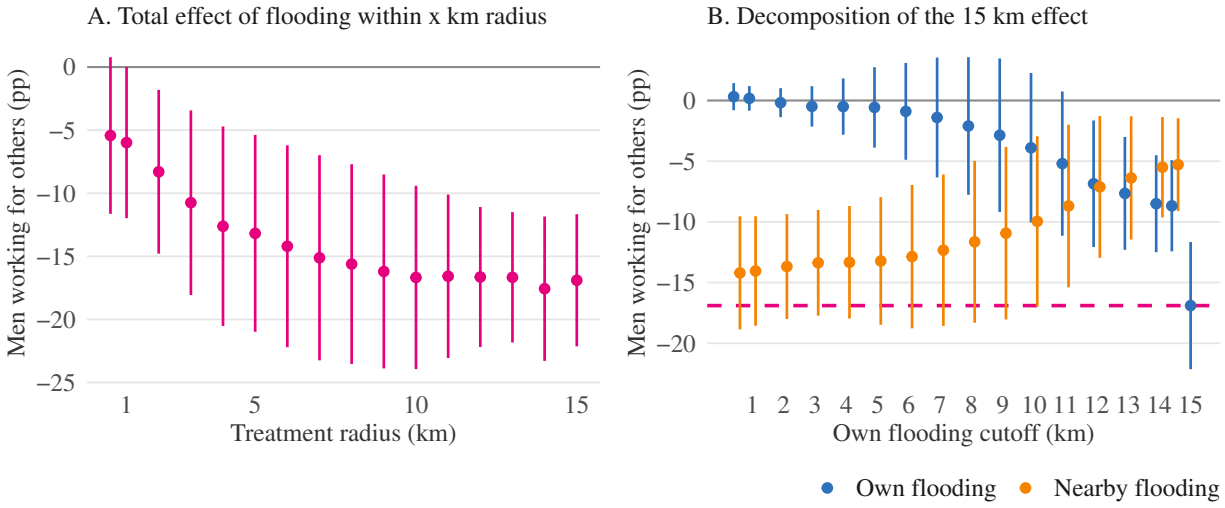


Figure 15: Effect of flooding on wage employment, agricultural households

Notes: Panel A plots the two-stage least squares coefficient on flooding within each radius, instrumented by the mean recentered hydrological prediction over the same radius, multiplied by the within-district difference between the 10th and 90th percentiles of that regressor. Panel B plots, for each cutoff, the coefficients on flooding within the cutoff (blue) and flooding from the cutoff to 15 km (orange) from one regression with both regressors, each instrumented by the recentered prediction over the same area. Each coefficient in Panel B is multiplied by the average change in its regressor when flooding within 15 km moves from its 10th to its 90th percentile within districts. The dashed line marks the 15 km estimate from Panel A, which is also the point at 15 km. The outcome is the share of the household's men aged 18 to 65 who work for others, controlling for the number of such men. In all panels the unit of observation is a household surveyed in both waves, and the sample is households with at least one man aged 18 to 65 whose pre-flood sector we observe, restricted to households whose employed men were all in agriculture before the floods (N = 1,830). Whiskers are 90% confidence intervals from Conley standard errors with a Bartlett kernel and a cutoff set from the instrument's spatial correlogram. All regressions include district fixed effects.

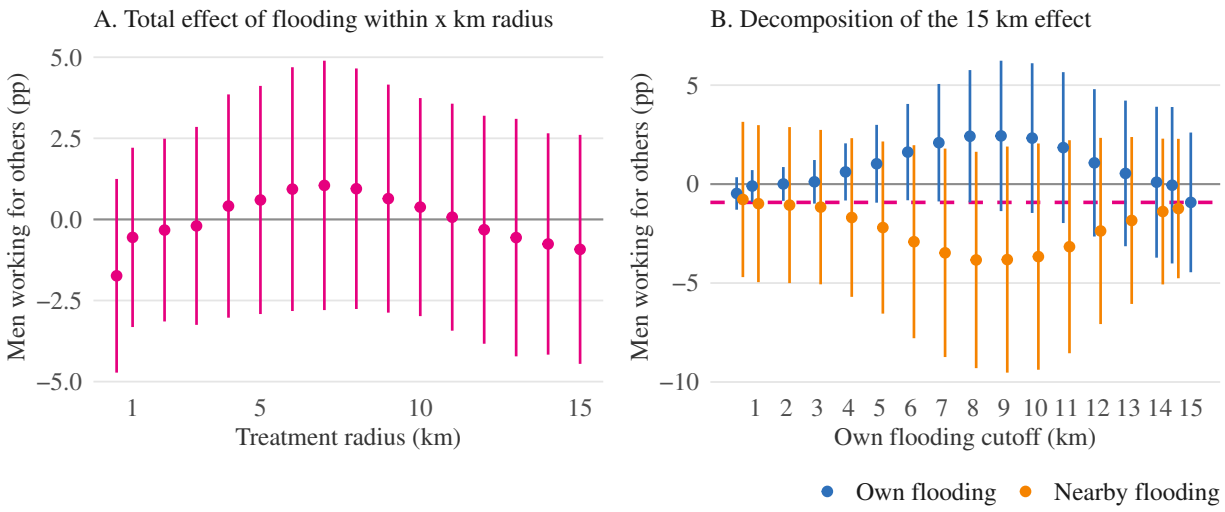


Figure 16: Effect of flooding on wage employment, non-agricultural households

Notes: Panel A plots the two-stage least squares coefficient on flooding within each radius, instrumented by the mean recentered hydrological prediction over the same radius, multiplied by the within-district difference between the 10th and 90th percentiles of that regressor. Panel B plots, for each cutoff, the coefficients on flooding within the cutoff (blue) and flooding from the cutoff to 15 km (orange) from one regression with both regressors, each instrumented by the recentered prediction over the same area. Each coefficient in Panel B is multiplied by the average change in its regressor when flooding within 15 km moves from its 10th to its 90th percentile within districts. The dashed line marks the 15 km estimate from Panel A, which is also the point at 15 km. The outcome is the share of the household's men aged 18 to 65 who work for others, controlling for the number of such men. In all panels the unit of observation is a household surveyed in both waves, and the sample is households with at least one man aged 18 to 65 whose pre-flood sector we observe, restricted to households with at least one man outside agriculture before the floods (N = 2,648). Whiskers are 90% confidence intervals from Conley standard errors with a Bartlett kernel and a cutoff set from the instrument's spatial correlogram. All regressions include district fixed effects.

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